Supporting Information: Substituting Reference σ-Profiles Degrades Solubility Prediction and Separates COSMO-SAC Error from Input Error

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This document is the Supporting Information for the article named above. Its own sections, tables, figures, equations and lemmas carry an S—§S1, Table S2, Fig. S1, Eq. (S1), Lemma S1—and a reference without one, such as §3.2 or Table 4, is to the article. Nothing here is a result the article does not state: every claim, every set it is measured on, its uncertainty and its scope are in the article's own text, and what this document adds is the proof, the per-row table, the robustness grid, the control and the negative behind each.

S1 Supplementary methods

COSMO-SAC constants. The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area $a_{\text{eff}} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\text{kcal}\,\text{Å}^4\,\text{mol}^{-1}\,\text{e}^{-2}$, hydrogen-bond coefficient $c_{\text{hb}} = 85580$ with cutoff $\sigma_{\text{hb}} = 0.0084\,\text{e}\,\text{Å}^{-2}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$; the σ-profile grid is 51 bins on $[-0.025, 0.025]\,\text{e}\,\text{Å}^{-2}$. The segment-activity fixed point is damped ($\lambda = 0.7$) and run for 16 iterations at train time and 30 at evaluation.

A usage pitfall in `cCOSMO` (`COSMO1` on typed profiles). Supplying the 2002 kernel with a typed three-profile base produces a silently-wrong result that the library returns without warning. The reference implementation[S1] pairs its 2002 model (`COSMO1`) with the single-profile Virginia Tech database; the demonstrated input is one untyped, Mullins-averaged profile. When `COSMO1` is instead supplied with a typed three-profile (Hsieh, `sigma3`) base — the configuration that arises when a 2002 and a 2010 arm share one profile source — it silently reads the nonpolar channel only, discards the donor and acceptor channels, keeps the total cavity area as prefactor, and returns a number without error. The prescribed way to recover a single 2002 profile from the typed set is instead their *sum*, $p = p_{\text{nhb}} + p_{\text{OH}} + p_{\text{OT}}$ (Eq. 9 of Ref.[S1]); the nonpolar-only read is neither that sum nor a documented mode. This single-channel arm scores MSE 1.263 on the VT-2005-matched set, whereas the *fair 2002* arm under the full convention (the same molecule supplied either to our differentiable layer or to `cCOSMO` with the full profile placed in the single nonpolar channel it actually reads) scores 1.757/1.758. This is an issue for the library maintainers (a silently-wrong return where an error would be safer), not a defect of any published benchmark. The kernel arms of §3.2 are deposited across two artifacts, and which file holds a value does not follow the set. The dispersion-*off* values (1.757 and 0.765 on the VT-2005-matched set, 1.911 and 1.978 on the broad IDAC set), all under the full convention in which the kernel comparison is run, are our differentiable layers on each kernel's native profiles (`fidelity_lever_fair2002.json`). Both dispersion values are `cCOSMO`'s own `COSMO3` call: the VT-2005-matched set's 0.785 from `closure_flip_ci.json`, where the reference returns no dispersion for the three bromine/iodine pairs and that arm therefore scores on n=57; the broad IDAC set's 1.855 from the `mse_2010dsp` field of `fidelity_lever_fair2002.json`, whose VT-2005-matched counterpart in the same file is instead our own layer's 7.43. That 7.43 is the measure of the gap: our transcribed dispersion term is not validated against the reference, which is why the dsp arm is deferred to `cCOSMO` throughout. The input-fixed control of §S3(a′) is a third file, `fidelity_lever_inputfixed.json`: it runs the 2002 kernel on the summed typed profile and re-derives the 1.757 and 0.765 arms of the first file as its own gate.

The differentiable COSMO-SAC-2010/dsp layer, and the scope of its dispersion caveat. The 2010 layer implements the four mechanisms of:[S2,S3] the donor/acceptor-typed three-profile representation, the temperature-dependent electrostatic misfit $c_{\mathrm{ES}}(T) = A_{\mathrm{ES}} + B_{\mathrm{ES}}/T^2$, the typed hydrogen-bonding kernel, and an optional dispersion term in the tabulated parameter ϵ/k_{B}. With dispersion *off* it reproduces the NIST COSMO-SAC-2010 reference to RMSE 1.4×10^{-4} $\ln\gamma$. That figure is quoted from the validation run's console output: unlike its 2002 counterpart, whose comparison is deposited as `closure_reference_validation.json`, the 2010 validator wrote no artifact, so this one figure cannot be checked against the deposited record. The dispersion term is not validated to that standard, for two reasons we state rather than absorb. It carries the tabulated coefficient at a single sign, the class-dependent sign reversal of Ref.[S3] being unimplemented and unexercised on the matched set, which holds no water or carboxylic-acid pair; and no ϵ is tabulated for the three matched pairs containing bromine or iodine, which our layer scores at ϵ=0. That alone accounts for the MSE 7.43 our layer records over all 60 VT-2005-matched pairs; on the remaining 57 it returns the reference 0.785, showing no divergence where both are defined. Every dispersion-off contrast in the paper is therefore computed on our layers and cross-checked against the NIST `cCOSMO` reference, and the *2010, dispersion on* runs are quoted from `cCOSMO` itself (§3.2).

Solver, curriculum and loss. The SLE fixed point is iterated as the damped map $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln\gamma_2(x_2^{(k)}, T)) + (1 - \lambda)x_2^{(k)}$ with closure-specific kernels. Gradients use the implicit-function theorem at the converged $x_2^\star$ (with $\eta = \partial \ln\gamma_2/\partial x_2$) rather than backpropagation through the iterations: exact at the fixed point, memory-constant in the iteration count, and well conditioned unless $1 + x_2^\star\eta \to 0$. The iteration counts above follow a convergence audit: a shorter n=8 schedule fails a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs. The three-phase curriculum is (P1) auxiliary-property pretraining on crystal, Hansen and $\ln\gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. Of the weighted loss, few terms are load-bearing for this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the σ-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\mathrm{fus}}$, on $\ln\gamma_2^\infty$ and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remainder — channel orthogonality, contrastive, mixture-of-experts balance, priors — is auxiliary. The grounded arms begin with a warm-up on the σ pool of up to 40 epochs, early-stopped on a held-out tenth, and the pinned recipe freezes the σ head for P2/P3, so that stream enters the warm-up and P1 only.

The closure itself, and the remaining terms of the 2002 layer. The COSMO-SAC layer of §2 maps a profile pair to $\ln\gamma_2$ through a damped fixed point for the segment activity coefficient Γ_S,

$$\Gamma_S(\sigma_m) = \Big[\textstyle\sum_n p_{\mathrm{norm}}(\sigma_n)\,\Gamma_S(\sigma_n)\,\exp\big(-\Delta W(\sigma_m,\sigma_n)/RT\big)\Big]^{-1}, \tag{S1}$$

with a precomputed exchange-energy matrix ΔW (electrostatic misfit plus a hydrogen-bonding term), a restoring contribution assembled from those segment activities, and an optional Staverman–Guggenheim combinatorial term. The restoring contribution is $\ln\gamma_2^{\mathrm{res}} = (A_2/a_{\mathrm{eff}})\sum_m p_2^{\mathrm{pure}}(\sigma_m)[\ln\Gamma_{S,\mathrm{mix}}(\sigma_m) - \ln\Gamma_{S,2}^{\mathrm{pure}}(\sigma_m)]$, and the optional Staverman–Guggenheim combinatorial term takes its size factor $r_i = V_i/r_0$ from the *reference* COSMO cavity volume so that it shares the cavity basis of the area A_i. Small hydride solvents such as water carry explicit hydrogens so that the polar and hydrogen-bond channels activate. The pair representation is passed through an MLP and may be augmented with RDKit descriptors, Morgan fingerprints and an ionic-context feature; the crystal head additionally offers Walden-entropy and group-contribution T_m modes. The two NRTL control closures of §2 are the Non-Random Two-Liquid model,[S4] the most expressive closure here, which fits interaction parameters per pair — predicting two interaction energies with a $1/T$ law and a non-randomness factor, optionally regularised by a UNIFAC group-contribution prior[S5] — and its symmetric one-parameter collapse $\ln\gamma_2(x_2,T) = \ln\gamma_2^{\infty}(T)(1-x_2)^2$, in which the identifiability analysis of §2.1 is cleanest. The crystal head's bounded parametrisations are below.

Crystal head. $T_m = 100 + 600\,\mathrm{logistic}(\cdot) \in [100, 700]\,\mathrm{K}$, $\Delta H_{\mathrm{fus}} = S_H\,\mathrm{softplus}(\cdot)$; the group-contribution T_m prior is affine- calibrated on the train split before entering the head.

Architecture. The headline COSMO-SAC model uses a message-passing encoder (a shared backbone with a 2-layer residual role-specific branch) of hidden width 64 over 3 message-passing layers, on 35-dimensional node and 8-dimensional edge features; a set2set readout (3 processing steps) pools each graph, solute and solvent interact through a 3-layer cross-attention block, and the 512-dimensional pair representation $[\,g_s, g_v, g_s \odot g_v, |g_s - g_v|\,]$ (dropout 0.012) feeds the crystal head, the 51-bin σ-profile head, and the parameter-free COSMO-SAC closure. The full model has 5.40M trainable parameters—capacity that the in-sample linear probe ($R^2 \approx 0.76$, §S6.1) indicates is not the binding constraint. The DirectGNN control shares this encoder, interaction, and readout verbatim, and adds a thermometer encoding of T in place of the temperature dependence the physics arm receives through $\Phi(T)$ and the closure. Figure S1 draws the two arms and marks where each of the two error terms enters.

Encoder backbones. The shared encoder has three interchangeable backbones: the headline message-passing network (MPNN); a hybrid local-message/global-attention graph transformer (GPS[S6]); and a physics-channelled variant, Thermodynamically-Informed Message Passing (TIMP), which splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward the Hansen parameters to prevent channel collapse. TIMP does not outperform the plain MPNN across our runs, consistent with the encoder being transfer-limited rather than expressivity-limited (linear probe train $R^2 \approx 0.76$, test $R^2 \approx 0.45$).

Optimisation, and the two arms' settings. The physics arm uses Adam (weight decay 5×10^{-4}, gradient clipping at 2.0, batch 64) under the three-phase curriculum of §2: 30 epochs of auxiliary pretraining (P1), 70 of full SLE training (P2), and 10 of low-rate stabilisation (P3), at learning rates 2.6×10^{-4}, 8.5×10^{-5}, and 8.5×10^{-6} respectively. The DirectGNN control was tuned separately and its settings differ: dropout 0.174 against 0.012, P2 learning rate 8.6×10^{-4} against 8.5×10^{-5},

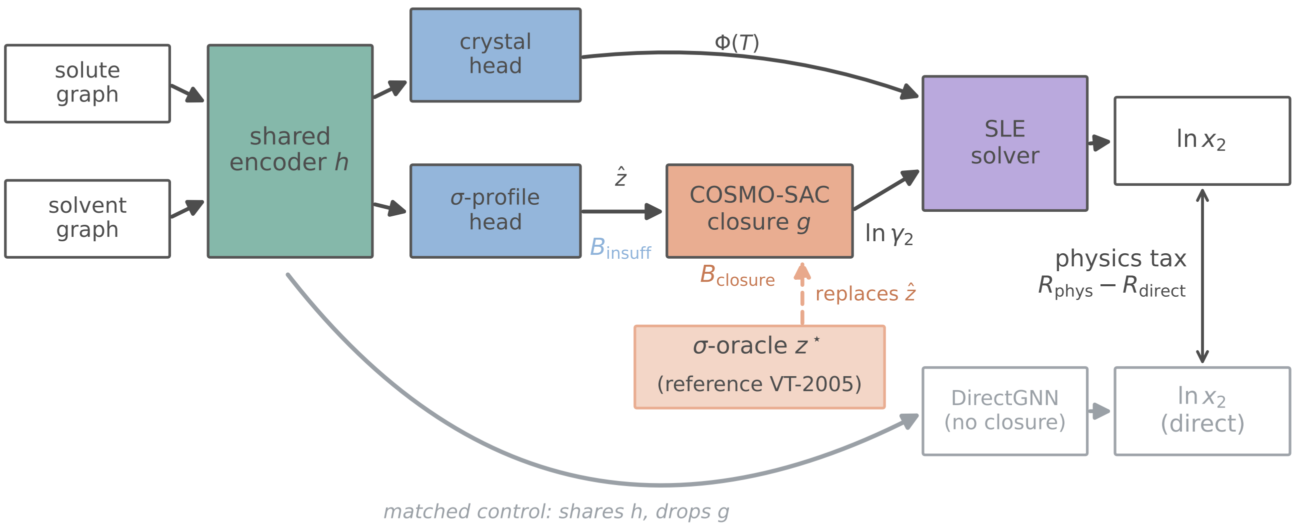

Figure S1: The model and the oracle substitution. Input insufficiency B_{insuff} enters at $\hat{z}$ and closure misspecification B_{closure} at the fixed map g; the σ-oracle (salmon) replaces $\hat{z}$ with the reference VT-2005 profile $z^\star$. DirectGNN (grey) is the capacity- and budget-matched control, separately tuned (§2.1), and the double-headed arrow between the two $\ln x_2$ outputs is the physics tax $R_{\text{phys}} - R_{\text{direct}}$.

weight decay 4.2×10^{-5} against 5×10^{-4}, gradient clipping 2.54 against 2.0, over 110 P2 epochs equal to the physics arm's total budget at the same width (64) and depth (3). Width, depth, batch size and epoch budget are therefore matched and the optimisation is not, so Δ_∞ and the physics-versus-control ΔMAE are differences between two separately tuned pipelines and are not attributable to the thermodynamic bottleneck alone (§2.1). A hyperparameter-matched control was not run.

Loss. The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hard-wired to 0 and two more (bridge, solubility variance) default to 0 through the configuration—disabled experimental scaffolding, released with the code for completeness. Table S1 lists the 13 carrying a non-zero default, together with the solubility-variance term and the σ-profile earth-mover term, which is applied outside the registry when the grounding stream is on; only the top two groups are load-bearing for the claims, and the *Loss-simplification ablation* paragraph below tests which.

Table S1: The 13 registry terms carrying a non-zero default weight, with the solubility-variance term (zero through the configuration) and the σ-profile earth-mover term, which is not a registry entry.[a] A further 17 registered terms are hard-wired to 0, and one more (bridge) is zero through the configuration.

role	term	default wt.
task	solubility (bin-weighted Huber on $\ln x_2$)	1.0
crystal grounding	melting point T_m	0.3
	fusion enthalpy ΔH_{fus}	0.3
σ/act. ground.	infinite-dilution $\ln \gamma_2^\infty$	0.5
	finite-composition $\ln \gamma_2$	0.5
	Hansen parameters	0.2
	σ-profile EMD (grounding stream on)	†
regulariser	solubility monotonicity in T	0.1
	physics preference	0.1
	adaptive-correction residual	0.01
	NRTL τ	0.01
	solubility variance	cfg
control coupling	DirectGNN NLL	0.2
	DirectGNN regulariser	0.05
architecture	mixture-of-experts balance	0.02

[a] † marks the σ-profile earth-mover term, active only when the grounding stream is on. Per-phase weight overrides are pinned in each run's resolved configuration.

Loss-simplification ablation. For a diagnostic result the elaborate objective is a potential confound: the paradox could be an artifact of loss tuning. The test registered in advance of the run trains the end-to-end grounded arm with only the task ($\ln x_2$ Huber), the crystal grounding ($T_m, \Delta H_{\text{fus}}$), and the σ-profile warmup, *dropping* the activity-grounding auxiliaries ($\ln \gamma_2^\infty$, $\ln \gamma_2$, Hansen) and every regulariser, and then reads the sign of the paradox (oracle vs. learned) and the surrogate drift. The registered prediction was that both persist, the mechanism being end-to-end training through a misspecified closure rather than the auxiliary terms, and both do at the level the ablation measures. The ablation re-measures the solubility *paradox* (reference minus learned $\ln x_2$ MAE), the observable effect, not the profile drift directly: on the full test set it is $+1.13$ (95% CI $[0.88, 1.38]$) under the minimal objective, *larger* than the $+0.30$ ($[0.20, 0.39]$) under the full one. This is the *overall*, solvent-dominated paradox, and it strengthens once the activity-grounding auxiliaries—the only pressure toward physics—are removed; on the both-reference subset (n=297 rows over the $\approx$7 VT-covered solutes) the sign reverses (-0.23 minimal vs $+0.22$ full), so the ablation is evidence that the overall observable effect is loss-robust, not subset-level confirmation of the mechanism. It is *consistent with* the drift being intrinsic to end-to-end training through a misspecified closure rather than a regularisation artifact. It is an implication of the amplified paradox, not a direct re-measurement of $\|\hat{\sigma} - \sigma\|$ under the minimal run (the minimal-loss checkpoint was not retained, and the larger reference-arm error is consistent with more profile drift but does not isolate it from closure-path recalibration). The minimal model trains normally (validation $\ln x_2$ MAE 1.87, within 0.1 of the full-loss base), ruling out a degenerate-training reading (seed 42; `scripts/analysis/run_loss_ablation_analysis.py`).

S2 Proofs

Lemma S1 (Structural non-identifiability of the SLE split)**.** *Consider the infinite-dilution SLE observable $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ with $\Delta C_p = 0$, so that $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ is affine in $u = 1/T$, and a symmetric two-suffix activity class ($\ln \gamma_2 = \ln \gamma_2^\infty (1-x_2)^2$) in which $\ln \gamma_2^\infty(T)$ is affine in u. Then, for any design with at least two distinct temperatures, μ depends on the parameters only through the two-dimensional statistic (its slope and intercept in u), and the activity intercept and slope already span that plane; hence the crystal parameters $(\Delta H_{\text{fus}}, T_m)$ are unidentified, the profiled Fisher information for ΔH_{fus} is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor $(1 - x_2)^2$ breaks the exact affinity, leaving a non-zero but near-degenerate information of order x_2^2.*

Proof of Lemma S1 (non-identifiability). Write $u = 1/T$ and $h = \Delta H_{\text{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\begin{aligned}\mu(u) &= -\Phi(u) - \ln \gamma_2^\infty(u) \\ &= -h(u - u_m) - (c + d\,u) \\ &= -(h + d)\,u + (hu_m - c),\end{aligned}$$

using $\ln \gamma_2^\infty(u) = c + d\,u$ (affine in u by hypothesis). Thus μ depends on the parameters $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$ only through the pair $(S, K) = (-(h + d),\ hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S, K)/\partial c = (0, -1)$ and $\partial(S, K)/\partial d = (-1, 0)$; hence for *any* crystal value (h, u_m) and any target (S, K) there is an activity pair $d = -S - h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h, u_m)\}$, so $(\Delta H_{\text{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S, K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out (c, d) annihilates it and $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$. At finite dilution $\ln \gamma_2(u) = \ln \gamma_2^\infty(u)\,(1 - x_2)^2$ adds $\ln \gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in u because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order x_2, so $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$. $\square$

Lemma S2 (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class $\mathcal{H}$ with tangent projection $\Pi_{\mathcal{H}}$, and let*

$v = \partial\mu/\partial\Delta H_{\text{fus}}$ be the crystal score. The efficient information for ΔH_{fus} is $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}}v\|^2$. If $\mathcal{H}$ contains the crystal score's span $\{1, 1/T\}$ (a free intercept and slope) then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma S1; if $\mathcal{H}$ is restricted (the slope is known or constrained) then $\mathcal{I}_{\text{eff}} > 0$ and the Cramér–Rao lower bound (CRLB) on ΔH_{fus} is finite. (The melting point T_m is likewise an unknown crystal parameter, but is profiled jointly: its score $\partial\mu/\partial T_m$ lies in $\operatorname{span}\{1\} \subset \operatorname{span}\{1, 1/T\}$*, already inside the nuisance tangent whenever the slope is free, so the dichotomy is unaffected by treating T_m as free.)*

Proof of Lemma S2 (class-dependent information). This is the semiparametric efficient-score construction[S7] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance σ^2, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\text{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the L^2-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}}v$ and the efficient information is $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}}v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u - u_m)/R \in \operatorname{span}\{1, u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \operatorname{span}\{1, u\}$ then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma S1; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing v (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \operatorname{span}\{1\}$) then $\Pi_{\mathcal{H}}v \neq v$ and $\mathcal{I}_{\text{eff}} > 0$, with a finite Cramér–Rao bound. □

Lemma 1, the identity this instrument rests on, is stated and proved in the article at §2.1.1: the cross term vanishes because $g(z^\star)$ is $\sigma(z^\star)$-measurable when $z^\star$ is the closure's whole argument, the Jensen bound needs no smoothness, the law-of-total-variance bound needs only that a binning of $g(z^\star)$ coarsens $\sigma(z^\star)$, and B_{insuff} is a functional of the law of $(m, z^\star)$ alone and therefore the same quantity under both combinatorial conventions—which is why Table S5 carries two valid upper bounds per set. Two consequences of that one-sidedness reconcile the bounding apparatus with the injectivity of the design. First, because every reported B_{insuff} is an *upper* bound and every B_{closure} the matching *lower* one, the bounds are conservative in one direction only: were the superpopulation variance negligible, the finding would be the stronger $B_{\text{closure}} = \text{MSE}$. Nothing reported can therefore be an artifact of having overestimated B_{insuff}, and no entry in any table is the share of the error the inputs contribute. Second, the resampling frequencies are statements about the *instrument* and not about the superpopulation: the clustered-bootstrap frequency P_{boot} is the fraction of cluster resamples in which the *measured bound* still falls below half the total error, MSE/2 being the threshold that orders the two terms. It measures whether this design resolves the ordering, not the probability that an inequality between two population quantities holds, so a low P_{boot} is loss of resolution and never evidence for the inputs.

Theorem S1 (Grounding-gap sandwich)**.** *Assume all risks are finite ($m, g(z^\star) \in L^2$; g and each $h \in \mathcal{E}$ measurable), (A1) the true encoder is representable, $\phi^\star \in \mathcal{E}$ with $z^\star = \phi^\star(x)$, and (A2) the intermediate is a lossless bottleneck, $\mathbb{E}[m \mid x] = \mathbb{E}[m \mid z^\star]$. Then*

$$\begin{gathered} 0 \;\le\; \mathcal{G} \;\le\; B_{\text{closure}}, \\ \mathcal{G} = B_{\text{closure}} - \big(R_{\text{e2e}} - R_{\text{free}}\big), \end{gathered} \tag{S2}$$

with $\mathcal{G} = B_{\text{closure}}$ iff $R_{\text{e2e}} = R_{\text{free}}$ (the composed class realizes the Bayes predictor by distorting the intermediate). In particular a well-specified closure ($B_{\text{closure}} = 0$) gives $\mathcal{G} = 0$; hence under A1–A2, *$\mathcal{G} > 0$ certifies $B_{\text{closure}} > 0$.*

Proof of Theorem S1 (grounding-gap sandwich). Throughout assume $m, g(z^\star) \in L^2$ and g, every $h \in \mathcal{E}$ measurable. *(a) $\mathcal{G} \ge 0$.* By A1, $\phi^\star \in \mathcal{E}$ is a feasible encoder with $g(\phi^\star(x)) = g(z^\star)$, so $R_{\text{e2e}} = \inf_h \mathbb{E}[(m - g(h(x)))^2] \le \mathbb{E}[(m - g(z^\star))^2] = R_{\text{orc}}$; the infimum need not be attained. Hence $\mathcal{G} = R_{\text{orc}} - R_{\text{e2e}} \ge 0$. *(b) $R_{\text{e2e}} \ge R_{\text{free}}$.* Every $g(h(x))$ is $\sigma(x)$-measurable, and $R_{\text{free}} = \mathbb{E}[(m - \mathbb{E}(m \mid x))^2]$ is the infimum of $\mathbb{E}[(m - p)^2]$ over *all* $\sigma(x)$-measurable p; an infimum over the subclass $\{g(h(\cdot)) : h \in \mathcal{E}\}$ is no smaller. *(c) $R_{\text{free}} = B_{\text{insuff}}$ under A2.* From $\operatorname{Var}(m) = \mathbb{E}[\operatorname{Var}(m \mid x)] + \operatorname{Var}(\mathbb{E}(m \mid x)) = \mathbb{E}[\operatorname{Var}(m \mid z^\star)] + \operatorname{Var}(\mathbb{E}(m \mid z^\star))$ and A2 ($\mathbb{E}(m \mid x) = \mathbb{E}(m \mid z^\star)$, so the two $\operatorname{Var}(\mathbb{E}(\cdot))$ terms agree), the two expected conditional variances agree: $R_{\text{free}} = \mathbb{E}[\operatorname{Var}(m \mid x)] = B_{\text{insuff}}$.

(d) Sandwich and identity. By Lemma 1, $R_{\mathrm{orc}} = B_{\mathrm{insuff}} + B_{\mathrm{closure}}$. With (c), $\mathcal{G} = R_{\mathrm{orc}} - R_{\mathrm{e2e}} = B_{\mathrm{closure}} - (R_{\mathrm{e2e}} - R_{\mathrm{free}})$, and (b) gives $R_{\mathrm{e2e}} - R_{\mathrm{free}} \geq 0$, so $0 \leq \mathcal{G} \leq B_{\mathrm{closure}}$; equality $\mathcal{G} = B_{\mathrm{closure}}$ holds iff $R_{\mathrm{e2e}} = R_{\mathrm{free}}$. Finally $B_{\mathrm{closure}} = 0$ forces $0 \leq \mathcal{G} \leq 0$, i.e. $\mathcal{G} = 0$; contrapositively, under A1–A2, $\mathcal{G} > 0 \Rightarrow B_{\mathrm{closure}} > 0$. *(e) A2 is necessary for the certificate.* If A2 fails, take $x \in \{a, b\}$ equiprobable, $z^\star \equiv 0$ (so A1 holds), $m(a) = +1, m(b) = -1$, and g with $g(0) = 0 = \mathbb{E}(m \mid z^\star)$ so $B_{\mathrm{closure}} = 0$; an encoder $h^\dagger(a) = 1, h^\dagger(b) = -1$ with $g(\pm 1) = \pm 1$ gives $R_{\mathrm{e2e}} = 0$ while $R_{\mathrm{orc}} = 1$, hence $\mathcal{G} = 1 > 0 = B_{\mathrm{closure}}$. Thus $\mathcal{G} > 0$ alone does not certify $B_{\mathrm{closure}} > 0$; the decomposition (Lemma 1) does. □

Proposition 1 is stated in the article at §2.1.2, with the asymptote-plus-variance form of the physics penalty $\mathcal{T}(n) \approx \Delta_\infty + [V_{\mathrm{phys}}(n) - V_{\mathrm{direct}}(n)]$ it is written against, and $\Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}$. It is proved here.

Proof of Proposition 1 (sign of the physics penalty). Write $D(n) := V_{\mathrm{direct}}(n) - V_{\mathrm{phys}}(n)$ and $\Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}$, so that $\mathcal{T}(n) = \Delta_\infty - D(n)$. *(i)* $\mathcal{T}(n) < 0$ exactly when $D(n) > \Delta_\infty$, immediately. *(ii)* If $\Delta_\infty > 0$ and $D(n) \to 0$, choose $n^\star$ with $|D(n)| < \Delta_\infty$ for all $n > n^\star$ (possible since $D(n) \to 0$); then $\mathcal{T}(n) = \Delta_\infty - D(n) > 0$. Monotonicity of D is *not* required for existence of $n^\star$; it is needed only for the stronger claim "$\mathcal{T}(n) > 0$ at every n," which we treat as an empirical observation. *(iii) A limiting refinement.* Since $V_{\mathrm{phys}}(n) \geq 0$, $D(n) \leq V_{\mathrm{direct}}(n)$: the variance a prior can save is bounded by the control's own estimation variance. As the black box approaches the aleatoric floor $V_{\mathrm{direct}}(n) \to 0$, so $\limsup_n D(n) \leq 0$; with both estimators consistent, so $V_{\mathrm{phys}}, V_{\mathrm{direct}} \to 0$, one has $D(n) \to 0$ and $\mathcal{T}(n) \to \Delta_\infty$. That limit is ≥ 0 when the control is asymptotically Bayes, so $R_{\mathrm{direct},\infty} = R_{\mathrm{free}}$ (the risk of an unconstrained predictor), and not merely from $V_{\mathrm{direct}} \to 0$; under that condition and A2, $\Delta_\infty = R_{\mathrm{e2e}} - R_{\mathrm{free}} = B_{\mathrm{closure}} - \mathcal{G}$ by (S2). □

The variance term $D(n)$ of Proposition 1 is what carries the semiparametric phenomenon §3.6 appeals to, in which an estimated propensity outperforms the known-true one:[S8,S9] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

S3 Per-arm tables and decomposition robustness

Table S2 gives the exact per-arm metrics behind the six three-seed bars of Fig. 2: three seeds on the cross-arm intersection lock (n=5608, rows supervised and finite in every arm); the two single-seed co-adaptation bars are valued in the paragraph below. Per-seed values are printed to three decimals; every mean, difference and ± quoted anywhere is computed on the unrounded values, so the printed triples reproduce them only to that precision. Every ± that spans *seeds* in this paper is a population standard deviation over them (`ddof`= 0); a sample standard deviation (`ddof`= 1) would be larger by $\sqrt{3/2}$, and the error bars of Fig. 2 follow the same `ddof`= 0 convention. Three kinds of ± in this paper are *not* that, and each says so where it is printed: the kernel-residual figures of the correctability test (46 ± 5%, −5 ± 16%, +2 ± 15%) carry a standard *error* over five fit seeds; a ± on a permutation or phase-randomised null (Table 4; §3.3) is that null's own spread over draws, not a spread over seeds at all; and the pKa arm errors of §S8 span the 20 within-scaffold splits. Reading the table off the rounded means mis-states differences by up to 0.01: the learned-σ and DirectGNN arms differ by 0.1435 from their unrounded means, which we quote as +0.14, while their rounded values 1.85 and 1.70 differ by 0.15. Both closure families—NRTL and COSMO-SAC over a learned σ-profile—trail the DirectGNN control (the physics penalty), though the arms are separately tuned and that gap is not attributable to the bottleneck (§3.6). The σ-oracle (reference VT-2005 profile) is the worst arm. Restoring the Staverman–Guggenheim combinatorial term makes the *grounded learned-σ* arm slightly worse (1.8457 → 1.8788)—the signature of

a misspecified map (§3.2); no oracle-plus-combinatorial arm was run, so the combinatorial contrast is against the grounded learned-σ row and not against the oracle row.
The unrounded means are 1.7022, 1.7950, 1.8457, 1.8788, 2.0434 and 2.2517, with population standard deviations 0.033, 0.071, 0.053, 0.091, 0.040 and 0.021. The two senses of grounding are the second and the last of these: training-time supervision moves the arm from 2.0434 to 1.8457 (helps, -0.20), evaluation-time substitution from 1.8457 to 2.2517 (hurts, $+0.41$).

σ-head schedule of these arms. The four COSMO-SAC rows come from three training runs under one pinned configuration — 30 P1, 70 P2 and 10 P3 epochs with `freeze_sigma_head_during_sle` on, so the σ head's weights are held from P2 onward. The two grounded rows precede that with a σ warm-up on the grounding pool (up to 40 epochs, early-stopped on a held-out tenth of the pool) and interleave the pool through P1; under the freeze the stream is inactive in P2/P3. The ungrounded row sets warm-up and stream budget to zero, so its σ head is shaped in P1 by the corpus-internal $\ln \gamma_2^{\infty}$ auxiliary loss alone, never by an external σ label, and is frozen from P2 in the same way. The σ-oracle row is the grounded checkpoint re-evaluated with $z^{\star}$ injected, so it shares the grounded schedule exactly. Because the freeze holds the head and not the encoder that feeds it, the predicted profile is not pinned at its warm-up value in these arms. We have not run the surrogate isolation (§3.3) on these checkpoints, so how far their latent drifted is not measured. The supervision contrast quoted there ($+0.40 \rightarrow +0.14$) is a separate matched pair of runs — one warm-up-plus-short-P1 base resumed twice for SLE, with the freeze off and on, on n=5571 matched rows — and not these arms.

Oracle-application controls: two of the three are single-seed. All three oracle applications (evaluation-only, training-time injection, channel-swap; §2) are computed on the same three-seed cross-arm row set as Table S2, and their per-arm predictions are released with the code. Only the evaluation-only substitution is a three-seed result. Training-time injection (MAE 1.98) and the channel swap (MAE 2.08) were run at seed 42 alone, so their matched comparator is the seed-42 learned-σ value 1.803 on the same n=5608 lock, not the three-seed mean 1.8457 of Table S2. Both exceed the worst of the three learned-σ seeds (1.921), so the direction of both survives a worst-case seed comparison. The channel-swap margin ($+0.27$) is 2.3 times the learned arm's full between-seed range; the training-time margin ($+0.18$) is one and a half times it, so that control fixes a sign rather than an effect size. This matters for how the paradox's magnitude should be read. The evaluation-only substitution ($+0.41$ against the three-seed mean) mixes closure misspecification with the ordinary degradation any pipeline suffers when an input distribution is swapped at test time; the training-time injection, in which the model co-adapts to the reference profile, is the arm that removes that component and it costs $+0.18$. The co-adapted arm is therefore the confound-free one and the evaluation-only $+0.41$ is not. How large a share of the gap co-adaptation carries off can be read only on the seed the two arms share: at seed 42 the injection costs $1.981 - 1.803 = 0.177$ where the evaluation-only substitution costs $2.232 - 1.803 = 0.428$, a share of 0.59. We report that share as a description of seed 42 and not as a magnitude, because it is not resolved. Its numerator was measured once, so the injected arm's own between-seed spread is unmeasured, and a spread the size of the learned arm's 0.118 would move the share by about ± 0.27; holding the injected arm at 1.981 and reading it against the seed-43 and seed-44 learned baselines instead gives 0.83 and 0.61. What survives that latitude is the sign—at the matched seed part of the evaluation-only gap does go with co-adaptation, which is what makes the co-adapted arm the confound-free one—and the sign, not the share, is what the claim rests on. The ratio $(0.406 - 0.177)/0.406 = 0.56$ sets the one-seed injection against the three-seed mean gap; it is a share of nothing and is reported nowhere. Neither $+0.18$ nor $+0.27$

Table S2: Per-arm scaffold-split metrics (3 seeds, n=5608 cross-arm lock), with per-seed MAE to three decimals. Ordered by MAE. The $\pm$ is a population standard deviation over the three seeds, computed on the unrounded values.

Arm	MAE	R^2	per-seed MAE (42/43/44)
DirectGNN (no closure)	1.70 ± 0.03	$+0.42 \pm 0.03$	1.749/1.674/1.684
NRTL closure	1.80 ± 0.07	$+0.34 \pm 0.03$	1.734/1.758/1.894
COSMO-SAC, learned σ (grounded)	1.85 ± 0.05	$+0.33 \pm 0.05$	1.803/1.921/1.813
+ Staverman–Guggenheim combinatorial	1.88 ± 0.09	$+0.32 \pm 0.05$	1.805/2.007/1.824
COSMO-SAC, ungrounded σ	2.04 ± 0.04	$+0.19 \pm 0.01$	2.040/1.996/2.093
COSMO-SAC, reference σ (σ-oracle)	2.25 ± 0.02	-0.03 ± 0.04	2.232/2.281/2.242

can be ordered against the -0.20 supervision gain either, whose own per-seed values are 0.237/0.075/0.281 and straddle $+0.18$.

The estimator grid. Three analyst choices set the separation margin on the VT-2005-matched set, and Table S3 reports every combination of them rather than one setting of each. The margin is positive in 66 of the 72 cells. The six exceptions are the coarsest conditionings—three bins under either combinatorial convention and four bins residual-only—where the law-of-total-variance bound is valid but weakest. The grid is not a set of interchangeable settings, because two opposing biases run across it. Coarser bins condition on less, so the bound is weaker; finer bins condition on more but leave few points per bin, and the within-bin variance is then downward-biased in finite samples, which inflates the margin favourably. Neither end is unbiased. We headline the eight-bin cell under the *unbiased* within-bin variance on an occupancy rule fixed in advance of the grid—the coarsest binning that still leaves fewer than eight rows per bin, here 7.5 at n=60—and we carry the same *bin count* to the broad IDAC set, so the two are conditioned at the same granularity. On 477 rows that is not the same occupancy rule (which would call for about 60 bins) and is deliberately conservative. The broad set's margin grows with bin count over the grid as a whole, though not cell by cell: in the deployed convention under the unbiased variance it falls from $+0.10$ at three bins to $+0.01$ at four, from $+0.57$ at seven to $+0.51$ at eight, and from $+0.83$ at twenty to $+0.82$ at twenty-four. The endpoints are what the choice turns on: $+0.51$ at eight bins against $+0.96$ at forty-eight. Eight bins therefore gives a looser bound and a smaller margin there than the occupancy rule would. It is *not* the least favourable cell of the middle range: seven bins residual-only under the unbiased variance gives $+0.01$ against the eight-bin $+0.05$, and twenty bins gives $+0.08$. Nothing turns on the choice—the whole middle range of the deployed convention lies between $+0.01$ and $+0.47$, all positive—but the reason for it is occupancy and not conservatism, and taking the least favourable middle cell instead gives $+0.01$ and the same verdict of an unresolved margin. The per-cell P holds that cell's bin count fixed across resamples. The main text instead quotes the deposited adaptive rule $\max(3, \lfloor n_{\mathrm{resample}}/10 \rfloor)$, which conditions on less on small resamples and is therefore more conservative: it gives $\approx$0.78 (maximum-likelihood) and $\approx$0.72 (unbiased) residual-only, against $\approx$0.89 and $\approx$0.86 full. The two schemes measure different things and we report both rather than the higher.

A fourth analyst choice: what "equal-count" means when n is not divisible by the bin count. The rule this paper deploys, and which §2.1.1 names as the fourth analyst choice without restating, is interpolated (`numpy` default `linear`) quantile edges of g at $j/8$, $j = 1, \dots, 7$, with a value falling exactly on an edge assigned to the upper bin. Neither n divides by eight ($60/8 = 7.5$, $477/8 = 59.625$), and that alone is what makes standard implementations of the same phrase disagree; there are no exact ties in g on either set under either convention, so each of them is a genuine coarsening of $\sigma(g)$ and the bound is valid under all of them. On the VT-2005-matched set at eight bins, the deployed convention and the unbiased variance, $B^{\mathrm{up}}_{\mathrm{insuff}}$ is 0.713 under our rule, under the same edges with ties sent to the lower bin, and under `pandas.qcut` alike; 0.624 under `numpy.array_split` of the g-sorted order (margin +0.225); 0.669 under $\lfloor k\,\mathrm{rank}/n \rfloor$ (+0.134); and 0.736 under order-statistic quantile edges (+0.001). Exact enumeration by dynamic programming over *every* partition of the g-ordered rows into eight contiguous bins of sizes in $\{7, 8\}$ puts the VT-2005-matched bound in $[0.620, 0.742]$ and its margin in $[-0.012, +0.232]$; at seven bins the margin envelope is $[-0.134, +0.157]$, and one such rule realises -0.089 there. The sign of the VT-2005-matched margin is therefore set by the binning rule and not by the data, which is a further reason to read its +0.05 as consistent with zero. The broad IDAC set absorbs the same choice: the six strictly equal-count implementations give 0.694 to 0.700, a spread of 0.006, and the exact envelope over all sizes-$\{59, 60\}$ layouts is $[0.687, 0.701]$, margin $[+0.499, +0.528]$ (at seven bins, $[+0.564, +0.570]$). Our 0.697 sits mid-range in both enumerations. Nothing load-bearing moves, since the VT-2005-matched set's margin is unresolved either way and the claim is carried on the broad set; we name the rule because the phrase "equal-count" does not by itself fix it.

The same grid on the broad IDAC set is Table S4, now with its own per-cell bootstrap frequency rather than a bare range of point margins. None of its 56 cells is negative. Within the *deployed* residual-only convention the margin runs from +0.01 (four bins) and +0.10 (three bins) up to +0.96, with the eight-bin unbiased headline at +0.51; the frequencies at the two coarsest cells, 0.65 and 0.42, say that those conditionings do not resolve the ordering, and they are why the broad set's aggregate margin is not resolved at every conditioning. Under the full convention the same grid runs from +0.23 to +1.25 and reaches $P > 0.95$ from five bins upward.

Table S3: Separation margin MSE $- 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the VT-2005-matched set (n=60), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] Negative margins in bold.

bins	full/ML	full/Be	res/ML	res/Be
3	**−0.06** (.64)	**−0.16** (.59)	**−0.43** (.43)	**−0.53** (.38)
4	+0.18 (.82)	+0.07 (.78)	**−0.14** (.63)	**−0.26** (.57)
5	+0.50 (.90)	+0.38 (.87)	+0.21 (.77)	+0.10 (.70)
6	+0.66 (.94)	+0.54 (.92)	+0.35 (.84)	+0.22 (.77)
7	+0.76 (.96)	+0.62 (.94)	+0.18 (.89)	+0.01 (.83)
8	+0.74 (.97)	+0.58 (.96)	+0.24 (.91)	+0.05 (.86)
9	+0.90 (.98)	+0.75 (.97)	+0.42 (.93)	+0.24 (.88)
10	+1.12 (.98)	+0.99 (.97)	+0.42 (.94)	+0.21 (.89)
11	+1.11 (.99)	+0.95 (.98)	+0.38 (.96)	+0.14 (.91)
12	+0.96 (.99)	+0.75 (.98)	+0.49 (.96)	+0.24 (.93)
13	+1.01 (.99)	+0.80 (.98)	+0.59 (.97)	+0.34 (.94)
14	+1.05 (.99)	+0.81 (.98)	+0.50 (.97)	+0.21 (.94)
15	+1.00 (.99)	+0.73 (.99)	+0.52 (.98)	+0.20 (.95)
16	+1.03 (1.0)	+0.77 (.99)	+0.56 (.98)	+0.22 (.96)
17	+1.25 (1.0)	+1.05 (.99)	+0.56 (.99)	+0.20 (.96)
18	+1.08 (1.0)	+0.76 (.99)	+0.69 (.99)	+0.39 (.97)
19	+1.11 (1.0)	+0.77 (1.0)	+0.79 (.99)	+0.47 (.97)
20	+1.23 (1.0)	+0.94 (.99)	+0.55 (.99)	+0.08 (.97)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive.

The decomposition on both sets, and the three kernel arms. Table S5 is the closure–insufficiency decomposition of §3.2.1–§3.2.2 on both sets under both conventions, with the maximum-likelihood variance variant, the constant-offset (Jensen) row and the bootstrap bands. Table S6 is the three COSMO-SAC arms of §3.2.3 on both sets under both conventions, with the bootstrap frequencies, the 90% margins and the block split of the 2002 → 2010 contrast into the 57 pairs shared with the VT-2005-matched set, the complementary 420 and the 80 of those at 298 K. Every claim either table carries has a row in Table 4 with its set, value and its uncertainty.

The bounds inventory, named and valued. Table S7 lists every bound either set supports at the aggregate level, together with the corroborating estimators Table S5 points here for instead of listing them row by row. The inclusion rule is note *a* of the table, and it is a rule for that table and not a claim of completeness over the paper: the LOTV half is a selection from two bin grids far too

Table S4: Separation margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ on the broad IDAC set (n=477), over bin count, within-bin variance (ML, Be) and combinatorial convention.[a] No cell is negative; the headline is the 8/res/Be cell.

bins	full/ML	full/Be	res/ML	res/Be
3	+0.24 (.60)	+0.23 (.59)	+0.11 (.42)	+0.10 (.42)
4	+0.50 (.93)	+0.49 (.93)	+0.02 (.67)	+0.01 (.65)
5	+0.84 (.98)	+0.82 (.98)	+0.50 (.77)	+0.48 (.76)
6	+0.84 (.99)	+0.82 (.99)	+0.51 (.86)	+0.50 (.85)
7	+0.94 (.99)	+0.93 (.99)	+0.59 (.89)	+0.57 (.88)
8	+0.96 (1.0)	+0.95 (1.0)	+0.53 (.93)	+0.51 (.92)
10	+1.06 (1.0)	+1.04 (1.0)	+0.67 (.96)	+0.64 (.94)
12	+1.05 (1.0)	+1.03 (1.0)	+0.74 (.97)	+0.71 (.97)
16	+1.11 (1.0)	+1.09 (1.0)	+0.76 (.99)	+0.72 (.99)
20	+1.17 (1.0)	+1.14 (1.0)	+0.88 (1.0)	+0.83 (.99)
24	+1.19 (1.0)	+1.15 (1.0)	+0.87 (1.0)	+0.82 (1.0)
32	+1.21 (1.0)	+1.16 (1.0)	+0.93 (1.0)	+0.86 (1.0)
40	+1.30 (1.0)	+1.25 (1.0)	+1.03 (1.0)	+0.95 (1.0)
48	+1.29 (1.0)	+1.22 (1.0)	+1.06 (1.0)	+0.96 (1.0)

[a] ML, maximum likelihood; Be, unbiased. In parentheses, the fixed-bin two-way cluster-bootstrap frequency that the margin is positive. Regenerates from `results/b_insuff/convention_audit.json`.

large to reproduce, and the $\ln\gamma_2^\infty$-stratified bounds of Table S13 are computed on subsets of the VT-2005-matched set rather than on either whole set and are not in it. It regenerates from `make_paradox_figures.py` (in `scripts/analysis/`) run with `--dump-bounds`, which prints the inventory, so the table can be re-checked against the artifacts. The regeneration's guard against an artifact changing under the inventory is *not* the same on the two sets, and we state which is which rather than claim one rule. On the VT-2005-matched set it stops both ways: on an estimator the artifact records and the inventory does not name, and on a named one the artifact no longer records. It additionally checks the entries that set's two audit artifacts share against each other. On the broad set it stops only the first way, so a named estimator absent from that set's audit artifact would be dropped rather than raised. $B^{\mathrm{up}}_{\mathrm{insuff}}$ and the B_{closure} lower bound it implies are contiguous shares of one total, so what separates the terms is the margin $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$ and not a gap between the two. Two rows of the table carry the whole risk on each set. On the broad set the coarsest and the least favourable cell differ—three bins gives 0.901, four bins 0.948 against a threshold of 0.951, a margin of +0.007 at bootstrap frequency 0.65 and the tightest anywhere on that set. On the VT-2005-matched set the two coincide at three bins (1.000), and both it and the four-bin cell (0.866) sit *above* that set's threshold of 0.736: the negative-margin cells Table S3 sets in bold, and both are listed here.

The stratification rule, in full. §3.2.1 reports the decomposition per stratum rather than per set, and the rule is set out here because it is the object a reader has to be able to check was not fitted to its own outcome.

Axis 1, solvent class. An ordered cascade over functional groups; the first class a solvent matches is the class it takes, and the order is hydrogen-bonding role first—the axis COSMO-SAC's H-bond term acts on—then dominant functional group. *(1) water* (111 rows / 15 pairs), the only liquid whose entire surface is H-bond active, with σ-profile mass at both extremes and the smallest cavity of any of them, so the Staverman–Guggenheim term is at its most extreme and the two combinatorial conventions disagree most. *(2) glycol ether*, at least one O–H and at least two oxygens: the ethylene-glycol oligomers (182 / 43), donor and acceptor on one flexible chain, so intramolecular H-bonds form and the profile is conformer-dependent—the documented worst case for a model that scores one gas-phase conformer as a bag of independent surface segments with no memory of connectivity. *(3) mono-alcohol*, exactly one O–H, exactly one O, no N, no halogen (35 / 17), the single-donor self-associating chemistry the 2002 H-bond term was fitted on. *(4) N–H protic* (33 / 33): formamide, N-methylformamide, primary and secondary amines, piperidine—nitrogen donors, weaker and differently shaped than O–H and lumped with them under one shared σ_{hb} threshold. *(5) aryl ether* (13 / 2), anisole and phenetole, tested *before* the general acceptor class so as not to be swallowed by it: lone pairs delocalised into the ring put the acceptor strength far below a dialkyl ether's, where a single cut-off is at its crudest. *(6) aprotic accep-*

Table S5: Closure–insufficiency decomposition on the reference-input activity path, as bounds rather than a point split. **The headline columns are the deployed residual-only ones on *both* sets**, one rule applied throughout; the full-convention columns are the comparison.

	broad IDAC (477)[a]		VT-2005-matched (60)[a]	
Quantity	**res**[b]	full[b]	**res**[b]	full[b]
$\mathrm{MSE}_{\mathrm{total}} = B_{\mathrm{closure}} + B_{\mathrm{insuff}}$ (exact)	1.90	1.91	1.47	1.80
B_{insuff} upper bound (LOTV)[c]	0.70	0.48	0.71	0.61
B_{closure} lower bound[d]	1.21	1.43	0.76	1.19
B_{closure} lower, constant-offset (Jensen)[e]	0.18	0.71	0.43	0.72
separation margin[f]	**+0.51**	+0.95	**+0.05**	+0.58
… bootstrap band[g]	$>0.95/\approx0.9/\approx0.9$	>0.95 (all three)	≈0.7	≈0.9
… maximum-likelihood variance[h]	0.68/+0.53	0.47/+0.96	0.62/+0.24	0.53/+0.74
… tighter of the two conventions[i]	0.48/+0.94		0.61/+0.26	
Further B_{insuff} upper bounds[j]	Table S7			
Label noise σ_η^2 (IDAC)	already inside every row above[k]			

[a] Broad IDAC, n=477 on UD profiles; VT-2005-matched, n=60 on VT-2005 profiles throughout (§3.2.3 scores the same pairs on UD). [b] *full*, with the Staverman–Guggenheim combinatorial term; *res*, residual-only. [c] Law of total variance, at eight equal-count bins of $g(z^\star, T)$ under the unbiased (Bessel-corrected) within-bin variance. It fits no model to the data. [d] $\mathrm{MSE} - B^{\mathrm{up}}_{\mathrm{insuff}}$; this row decides the ordering. [e] $(\mathbb{E}[m] - \mathbb{E}[g])^2$. [f] $\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$, the quantity that orders the two terms. [g] Pair / two-way / source clusters. The VT-2005-matched set supports the two-way clustering only, so it carries one entry per column. [h] Same rows and bins, maximum-likelihood within-bin variance, as B_{insuff} upper / margin. [i] Charged against the deployed MSE, as B_{insuff} upper / margin. Not headlined. [j] The entries of Table S7: out-of-fold RF and ridge (random and blocked folds), kNN-in-$z^\star$, and the binning-free $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})]$. Not a paper-wide inventory: the bin grids behind the LOTV selection are Tables S3 and S4, and the $\ln\gamma_2^\infty$-stratified bounds, computed on VT-2005-matched subsets, are Table S13. [k] A *component* of B_{insuff}, not a term beside it (§2.1.1).

tor (57 / 57): carbonyl, nitrile, azine nitrogen, ether oxygen, nitro or tertiary-amine nitrogen, with no donor—pentyl acetate, pyridine and four methylpyridines, acetone, acetonitrile. *(7) halogenated* (30 / 7), any C–F/Cl/Br/I and no donor: large soft polarisable surfaces against a kernel with no dispersion term and one effective contact area. *(8) aromatic hydrocarbon* (14 / 9), a quadrupolar π face with σ near zero over the ring, and *(9) aliphatic hydrocarbon* (2 / 2), pure dispersion, $\sigma \approx 0$ everywhere, so the residual term nearly vanishes and the prediction is almost entirely combinatorial. The nine are exhaustive and disjoint over the set's 36 solvents. Ether, nitro and tertiary-amine tests sit in the cascade because the same cascade is applied to the *solutes*, where those groups occur; no solvent reaches them.

Two further granularities, so that the granularity is not itself a choice. Coarse: donor-and-acceptor oxygen (classes 1–3, n=328), donor nitrogen (4, 33), acceptor-only (5–6, 70), halogenated non-H-bonding (7, 30), hydrocarbon non-H-bonding (8–9, 16). Fine: amide N–H 25 against amine N–H 8; ester 30, azine 23, ketone or aldehyde 2, nitrile 2; halogenated aliphatic 18 against halogenated aromatic 12. All three levels are in the deposited table, and two of their cells are worth naming. The coarse *acceptor-only* class, which merges the aryl ethers into the aprotic acceptors, carries a *positive* margin (+1.43 deployed at $P \approx 0.7$) where its dominant child is negative—the composition effect of §3.2.1 reappearing one level up, on 13 rows. At the fine level the azine sub-family changes sign between conventions at n=23. Neither overturns a headline-level verdict, and both are why the granularity is reported rather than chosen.

Axis 2, solute role. Water as the dilute solute (37 / 6) against organic solutes (440 / 179), reported alone and crossed with axis 1. Water as the solute is a different physical situation from water as the solvent: an isolated H_2O in an organic continuum has all four H-bond sites unsatisfied by like molecules, a state the

Table S6: Activity error (MSE, $\ln \gamma_2^\infty$ units) of the three COSMO-SAC arms, and the block split that reads the 2002 → 2010 reversal against profile database and temperature held fixed.

	VT-2005-matched (60), UD[b]		broad IDAC (477), UD[b]	
(a) Arm[a]	full[b]	res[b]	full[b]	res[b]
fair 2002	1.757	1.436	1.911	1.902
2010, dispersion off	0.765	0.831	1.978	2.893
2010, dispersion on	0.785[e]	—[d]	1.855	—
P (2010 better)[c], dispersion off	>0.95	—	≈0.6	≈0.2
90% margin[c]	[+0.06, +2.18]	—	[−1.95, +0.93]	[−3.06, +0.70]
P (2010 better)[c], dispersion on	—	—	≈0.6	—
90% margin[c]	—	—	[−2.24, +1.31]	—

(b) Block (UD profiles, full convention)	2002 → 2010, dispersion off	P^c	90% margin[c]
shared with the VT-2005-matched set (57)	1.783 → 0.767	>0.95	[+0.06, +2.36]
complementary (420)	1.928 → 2.143	≈0.5	[−2.41, +0.87]
… of these, at 298 K (80)	0.796 → 1.284	≈0.3	—

[a] The three arms, named here and in the text: *fair 2002*, COSMO-SAC-2002 in our NIST-validated layer reading all three profile channels; *2010, dispersion off* and *2010, dispersion on*, COSMO-SAC-2010 without and with the dispersion term (§3.2.3). [b] *full* / *res* as in Table S5. *Both* sets are scored on UD profiles here, the VT-2005-matched set included—Table S5 scores the VT-2005-matched set on VT-2005 instead, and that database difference is the whole difference between its VT-2005/full 1.80 and the 1.757 here (§3.2.3). Within a set, each arm uses the σ-profile averaging its own model prescribes (Mullins for 2002, Hsieh for 2010). [c] P is the two-way cluster-bootstrap frequency that the 2010 arm is the better one, with the 90% margin on the paired squared-error difference beneath it. [d] A dash marks a cell the deposited artifacts do not carry. [e] This one cell is n=57, not 60: `cCOSMO` tabulates no dispersion ϵ for the three bromine- or iodine-containing VT-2005-matched pairs. On those common 57 the *dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n, and no claim rests on that ordering (§3.2.3).

segment-activity expression is not fitted near and the one COSMO-SAC-2002's documented weakness is specific to. The same cascade applied to the solute gives a secondary solute-family axis, also deposited, because a practitioner deciding where to ground needs both halves of a pair.

Two mechanical points. `classify_solvent` takes one argument, a SMILES string, and the whole cascade is a sequence of SMARTS tests: it never reads m, g, the squared error, MSE, B_{insuff}, the margin, the source DOI or the bootstrap. That is a property of the code and not an assurance. Second, the University of Delaware compound list stores formamide and N-methylformamide as imidic-acid tautomers (`N=CO`, `CN=CO`); RDKit's canonical tautomer restores the amide so that the N–H donor defining their class is visible, and that normalisation is applied to every molecule alike rather than carved for two of them.

What sets the level of the bound, and why it barely moves between strata. From above, $B^{\text{up}}_{\text{insuff}}$ is capped by the within-stratum label variance: the law of total variance on a single bin *is* $\mathrm{Var}(m)$, and refining the bins can only lower it, so under the maximum-likelihood within-bin variance $B^{\text{up}}_{\text{insuff}} \leq \mathrm{Var}(m)$ identically—which the deposited `b_insuff_up_ml` column satisfies in every stratum (0.12 against 0.80 on the aprotic acceptors, 0.10 against 3.07 on the glycol ethers, 0.20 against 1.42 on water, 0.22 against 5.61 on the N–H protics, 0.44 against 3.66 on the halogenated, 0.58 against 0.60 on the mono-alcohols). The Bessel-corrected estimate the manuscript headlines can exceed $\mathrm{Var}(m)$ once bins hold only a few rows, and does so exactly once: the mono-alcohols, 0.733 against 0.604 at n=35 and four rows a bin, where the artifact raises `b_insuff_exceeds_var_m`. That flag fires only on a stratum already declared not boundable, which is the boundability crite-

Table S7: **Upper bounds on the input-insufficiency term** B_{insuff}, on both sets taken whole—every bound the inclusion rule keeps, plus one estimator that set cannot support, with the value, what each fits and what qualifies it. What is included, and what is not: note *a*. The per-stratum bounds that replace the aggregate margin are Tables S8–S12 below, drawn as Fig. S2.

Estimator[a]	$B^{\text{up}}_{\text{insuff}}$	fits[b]	note[c]
Broad IDAC set (n=477, UD profiles). MSE = 1.902; threshold MSE/2 = 0.951; headline margin +0.51.			
LOTV, 8 bins, unbiased variance [**headline**]	0.697	no model (○)	
LOTV, 8 bins, maximum-likelihood variance	0.685	no model (○)	
LOTV, 3 bins (coarsest)	0.901	no model (○)	
LOTV, 4 bins (least favourable)	0.948	no model (○)	tightest margin on this set
LOTV, 8 bins, full convention (also valid)[d]	0.482	no model (○)	
RF out-of-fold, folds blocked on the molecule pair	0.882	out-of-fold (◇)	
ridge out-of-fold, folds blocked on the molecule pair	1.467	out-of-fold (◇)	above thr.
RF out-of-fold, folds blocked on the pair, conditioning on $(z^\star, T)$	0.823	out-of-fold (◇)	
RF out-of-fold, folds blocked on the solute	2.043	out-of-fold (◇)	vacuous
RF out-of-fold, random folds	0.146	out-of-fold (◇)	leaky
ridge out-of-fold, random folds	0.801	out-of-fold (◇)	leaky
$\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$, binning-free, on the 359 rows that support it	0.046	no model (○)	single-source
VT-2005-matched (n=60, VT-2005 profiles). MSE = 1.472; threshold MSE/2 = 0.736; headline margin +0.05.			
LOTV, 8 bins, unbiased variance [**headline**]	0.713	no model (○)	
LOTV, 8 bins, maximum-likelihood variance	0.618	no model (○)	
LOTV, 3 bins (coarsest, least favourable)	1.000	no model (○)	above thr.
LOTV, 4 bins	0.866	no model (○)	above thr.
LOTV, 8 bins, full convention (also valid)[d]	0.608	no model (○)	
kNN-in-$z^\star$ (convention-independent; least conservative)	0.400	out-of-fold (◇)	deflated
kNN, neighbour from a distinct solute	0.448	out-of-fold (◇)	
kNN, neighbour from a distinct solvent	1.458	out-of-fold (◇)	above thr.
kNN, neighbour from a distinct pair	1.831	out-of-fold (◇)	vacuous
RF out-of-fold, random folds	0.621	out-of-fold (◇)	deflated
ridge out-of-fold, random folds	0.625	out-of-fold (◇)	
RF out-of-fold, folds blocked on both molecules	3.653	out-of-fold (◇)	vacuous
ridge out-of-fold, folds blocked on both molecules	divergent	out-of-fold (◇)	
ridge out-of-fold, deliberately aggressive	0.877	out-of-fold (◇)	above thr.
$\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$[e]	—	—	no repeated pairs

[a] One inclusion rule governs both sets, in two halves that differ in kind. The LOTV half is a *selection* from bin grids too large to draw (56 cells on the broad set, 72 on the VT-2005-matched set; Tables S4 and S3): *within the deployed convention and under the unbiased within-bin variance*, the headline cell, the *coarsest*, the *least favourable* and every such cell above the threshold, with the headline cell's maximum-likelihood variant and its full-convention counterpart listed beside it. Cells above the threshold under the other variance or the other convention are not listed; they are in those two grids, where Table S3 sets the negative-margin ones in bold. The estimator half is every out-of-fold estimator that set's audit artifact records, in every fold design it records, vacuous designs included—the per-set guards on that are stated with the regeneration command above. [b] What the estimator fits to the data. *no model* (○) for the law-of-total-variance (LOTV) cells and for $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$, *out-of-fold* (◇) for a fitted estimator. The LOTV bound, fitting nothing, is unaffected by any fold design. [c] *above thr.*, the bound exceeds that set's separation threshold MSE/2, so that cell does not separate; *vacuous*, it exceeds the total MSE, which two of the VT-2005-matched set's estimators and one of the broad set's do; *leaky*, random folds leave a held-out row's own molecule pair in the training fold at a neighbouring temperature; *deflated*, biased low by the crossed design, in which two pairs sharing a molecule agree on 51 of the 102 $z^\star$ coordinates; *single-source*, every multi-row pair supporting the bound comes from one publication, so it is valid on those rows but samples no between-laboratory component and is not promoted. [d] Computed under the other combinatorial rule and reported against the deployed-convention threshold beside it. It is valid there, and it is the only entry on either set that is not residual-only. [e] The estimator that set cannot support.

rion doing its job rather than an unexplained anomaly—and that same value is what sets the $\times 6.8$ span quoted in §3.2.1.

From below the received story needs a correction. $z^\star$ is a deterministic function of the pair, so $\mathbb{E}[\mathrm{Var}(m \mid \text{pair}, T)]$ *is* B_{insuff} on rows that repeat; but no two of the 477 share a pair *and* a temperature, so pure replicate disagreement is not measurable on this set. The coarser $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$, which additionally absorbs the whole within-pair temperature spread, is 0.0003 on the mono-alcohols, 0.037 on water, 0.044 on the glycol ethers, 0.075 on the halogenated and 0.046 over the 359 rows that support it on the whole set—one to two orders of magnitude below the binning bound of 0.11–0.73. The level of $B^{\text{up}}_{\text{insuff}}$ here is therefore not set by label noise but by the residual spread of m inside a quantile bin of a scalar, which is a property of the estimator rather than of the chemistry. That is the sharper form of the structural claim, and the reason the bound is so nearly stratum-independent while MSE is free to move by a factor of 25 to 51.

The structural claim as a test. 2000 random partitions of the 477 rows into blocks of exactly the observed class sizes, at the same estimator cell and under the same boundability filter. Deployed convention: observed MSE range 2.163 against a null median of 0.405, no draw reaching it; observed $B^{\text{up}}_{\text{insuff}}$ range 0.107 against 0.165, $p = 0.26$; observed ratio 20.2 against 2.46, $p = 0.02$. Full convention: 2.353 against 0.337, again no draw reaching it; 0.058 against 0.133, $p = 0.12$; ratio 40.5 against 2.48, $p = 0.003$. The chemical partition separates the error far beyond chance and separates the bound no more than chance, which is what makes an aggregate margin a composition statistic.

The leave-one-out curves. At the headline cell under the deployed convention, baseline +0.509: leaving out `10.1021/acs.jced.7b00114` gives +0.184 and leaving out `10.1021/je020196d` (105 rows) +0.419, while the remaining thirteen deletions run from +0.503 to +0.875. Under the full convention the same curve is flat, $[+0.821, +1.073]$ against a baseline of +0.947, so the single-publication dependence is specific to the deployed convention. All 185 leave-one-pair-out margins lie in $[+0.422, +0.721]$ with median +0.507 and none non-positive, the seven most damaging deletions being *five* of the six water-as-solute pairs interleaved with two glycol-ether ones; the sixth water pair ranks 178th of 185, so those six rows do not move the aggregate in one direction. Leave-one-solvent-class-out spans $[+0.352, +1.156]$—removing water raises it most, by +0.647, and removing the halogenated solvents lowers it most, by −0.157—so exactly one class of the nine moves the aggregate by more than the aggregate's own distance from zero, and six of the other eight move it by less than 0.38. The span is wide because one class is; it is not evidence that the aggregate is broadly composition-sensitive, and the composition argument of §3.2.1 rests on the compounded cut and the null-partition test rather than on this curve. Per-stratum leave-one-source-out, and the VT-2005-matched set's own curves, are in the artifact and summarised in §3.2.1–§3.2.2.

S3.1 The map, stratum by stratum

What is measured and not established. Water carried as the dilute solute is 37 rows and six pairs from `10.1021/acs.jced.7b00114` alone, which supplies 100% of its rows and 100% of its squared error: at n=37 it is not boundable, and with one publication there is no deletion to run, so its +6.33 was never put at risk and is not evidence about water solutes. The halogenated solvents (30 rows, +3.39) are not boundable either, and 83% of that cell's squared error is the same publication; deleting it leaves 12 rows, fewer than the fixed cell needs, so the sign cannot be verified. Neither may be used to explain a reversal. The coarse acceptor-only class, whose +1.43 this table names, is 96% carried by that publication's 13 aryl-ether rows and goes to −0.19 when it is removed. Water as a *solvent* (+0.05 at $P \approx 0.8$ deployed, +0.57 full) flips under two of its three deletions; the coarse donor-and-acceptor-oxygen class—"the hydroxylic solvents"—flips to −0.42 when

Table S8: **The map, solvent axis, in full**: every solvent stratum of the broad IDAC set, including the ones the estimator cannot bound, at the cell §2.5 fixes—eight equal-count bins of $g(z^\star)$, unbiased within-bin variance, row unit, deployed residual-only convention—applied identically to all of them. *full* is the same cell under the full combinatorial convention. The map runs over 59 strata in all: the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole, over 477 rows, 185 pairs, 78 solutes, 36 solvents and 15 source publications, the finer solvent granularity in Table S9 and the solute axis in Tables S10 and S11. These four tables carry all 59, so nothing is dropped for being small, and each row states either what it shows or the test it fails; the article states the one row set the admissibility rule establishes, and which chemistry it does not.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent class								
glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.27, +2.83]	+2.23	4	**the closure's own error exceeds the bound on its inputs**, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
aprotic acceptor	57	57	4 (P7, 0.32)	−0.19	[−0.44, +0.12]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
mono-alcohol	35	17	2 (P3, 0.55)	+0.78	[+0.18, +3.24]	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
N–H protic	33	33	3 (P11, 0.62)	+0.61	[−0.27, +1.96]	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated	30	7	2 (P13, 0.83)	+3.39	[+0.86, +8.12]	+1.95	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there
aromatic hydrocarbon	14	9	2 (P13, 0.88)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aryl ether	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
Solvent class, coarse								
H-bond donor and acceptor (O)	328	75	8 (P8, 0.40)	+0.53	[−0.21, +2.18]	+1.34	3	the margin changes sign when one publication is deleted: P15 takes +0.527 → −0.417
H-bond acceptor only	70	59	5 (P13, 0.96)	+1.43	[−0.28, +6.19]	+0.77	0	the margin changes sign when one publication is deleted: P13 takes +1.434 → −0.192
H-bond donor (N)	33	33	3 (P11, 0.62)	+0.61	[−0.30, +2.11]	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated, non-H-bonding	30	7	2 (P13, 0.83)	+3.39	[+1.10, +8.61]	+1.95	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P13 (1 of 2 deletions), so the sign is untested there
hydrocarbon, non-H-bonding	16	11	3 (P13, 0.88)	+2.99	[−0.07, +8.86]	+1.84	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P10, P11, P13 (3 of 3 deletions), so the sign is untested there

[a] Number of source publications, then the largest one and its share of that stratum's squared error. The keys, used in all five map tables: P1 `10.1016/j.fluid.2013.06.028`, P2 `10.1016/j.fluid.2015.01.010`, P3 `10.1016/j.fluid.2015.11.037`, P4 `10.1016/j.fluid.2016.08.030`, P5 `10.1016/j.fluid.2016.10.009`, P6 `10.1016/j.jct.2013.05.006`, P7 `10.1016/j.jct.2015.12.034`, P8 `10.1016/j.jct.2016.10.013`, P9 `10.1016/j.jct.2018.05.003`, P10 `10.1016/j.jct.2019.05.011`, P11 `10.1016/j.tca.2012.02.005`, P12 `10.1016/j.tca.2015.11.010`, P13 `10.1021/acs.jced.7b00114`, P14 `10.1021/acs.jced.9b00726`, P15 `10.1021/je020196d`. [b] Two-way solute × solvent cluster bootstrap, 3000 draws, redrawn at each granularity, so one row set's margin repeats exactly while its endpoints differ in the last digits. [c] In how many of the four unit × convention cells the stratum is both boundable ($n \geq 40$, so eight bins hold five rows) and keeps its margin's sign under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 strata—three distinct row sets—satisfy. [d] The condition the row is in, not a label for it. A one-source cell cannot be tested and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive establishes $B_{\text{closure}} > B_{\text{insuff}}$ there, non-positive is failure to separate and licenses nothing (Table 4, note *i*).

Table S9: **The map, solvent axis at the fine granularity**—the third granularity of Table S8, at the same estimator cell and under the same rule, whose notes apply here unchanged. A finer class is a subset of a coarser one, so these rows are the same measurements re-partitioned and not new ones.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent family, fine								
glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.27, +2.87]	+2.23	4	**the closure's own error exceeds the bound on its inputs**, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.16, +0.40]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
mono-alcohol	35	17	2 (P3, 0.55)	+0.78	[+0.29, +3.26]	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
ester	30	30	1 (P9, 1.00)	−0.06	[−0.08, +0.03]	−0.05	0	$n < 40$: too few rows for eight bins of five; one source (P9), so there is no deletion to run and the sign was never put at risk
amide N–H	25	25	2 (P12, 0.96)	−0.24	[−0.34, +0.76]	−0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 2 deletions), so the sign is untested there
azine	23	23	2 (P11, 0.58)	−0.05	[−0.15, +0.14]	+0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P5 (1 of 2 deletions), so the sign is untested there
halogenated aliphatic	18	5	2 (P3, 0.87)	+0.11	[−0.18, +2.33]	−0.02	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3, P13 (2 of 2 deletions), so the sign is untested there
aromatic hydrocarbon	14	9	2 (P13, 0.88)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aryl ether	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated aromatic	12	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
amine N–H	8	8	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ketone/aldehyde	2	2	1 (P7, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitrile	2	2	1 (P7, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n

Table S10: **The map, solute axis**—the solute half of Table S8, at the same estimator cell and under the same rule, whose notes apply here unchanged. Rows at different granularities, and rows on the two axes, can be the same measurements: 362 of the 1711 stratum pairs share rows, and every row that keeps its sign carries its overlap with the others.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solute role								
organic solute	440	179	14 (P8, 0.35)	+0.18	[−0.23, +1.36]	+0.82	2	the margin changes sign when one publication is deleted: P15 takes +0.184 → −0.094
water as solute	37	6	1 (P13, 1.00)	+6.33	[+3.97, +9.49]	+3.54	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
Solute family								
alkane	129	45	7 (P15, 0.53)	+1.38	[+0.79, +2.33]	+1.72	4	the sign holds under every deletion, but 84% of these rows are the glycol ethers' and the 21 outside give −0.22: these are those rows again; 10% shared with the aprotic acceptors
aromatic hydrocarbon	71	32	7 (P15, 0.53)	−0.19	[−1.01, +0.75]	−0.07	0	the margin changes sign when one publication is deleted: P6 takes −0.188 → +0.081
ketone	63	11	3 (P14, 1.00)	−0.53	[−0.85, +0.64]	−0.33	0	too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there
mono-alcohol	39	21	5 (P11, 0.46)	+0.27	[+0.09, +2.10]	+0.44	0	$n < 40$: too few rows for eight bins of five; the sign holds, but all 5 deletions leave $n < 40$
water	37	6	1 (P13, 1.00)	+6.33	[+3.39, +9.50]	+3.54	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
alkene	32	16	4 (P8, 0.98)	+2.29	[+0.12, +3.43]	+2.76	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P8 (1 of 4 deletions), so the sign is untested there
aldehyde	29	3	1 (P2, 1.00)	−0.20	[−0.51, +0.13]	+0.68	0	$n < 40$: too few rows for eight bins of five; one source (P2), so there is no deletion to run and the sign was never put at risk
halogenated aliphatic	27	9	3 (P3, 0.99)	+1.06	[+0.15, +1.99]	+1.36	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 3 deletions), so the sign is untested there
amine N–H	12	12	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ether	12	4	3 (P2, 0.98)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated aromatic	8	8	4 (P1, 0.75)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
azine	6	6	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
tertiary amine	4	4	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
ester	3	3	2 (P5, 0.56)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
alkyne	2	2	2 (P5, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitrile	2	2	2 (P9, 0.98)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
nitro	1	1	1 (P9, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n

Table S11: **The map, the solvent × solute cross-tab and the set as a whole**—the last two granularities of Table S8, at the same estimator cell and under the same rule, whose notes apply here unchanged. The whole-set row is the aggregate the article declines to state, printed with the rest; the reading below states which of the cells cited together overlap and by how much.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent class × solute role								
glycol ether | organic solute	182	43	3 (P8, 0.53)	+2.04	[+1.23, +2.90]	+2.23	4	**the closure's own error exceeds the bound on its inputs**, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water | organic solute	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39]	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
aprotic acceptor | organic solute	57	57	4 (P7, 0.32)	−0.19	[−0.40, +0.12]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
mono-alcohol | organic solute	35	17	2 (P3, 0.55)	+0.78	[+0.30, +3.37]	+1.09	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P3 (1 of 2 deletions), so the sign is untested there
N–H protic | organic solute	33	33	3 (P11, 0.62)	+0.61	[−0.23, +1.99]	+0.86	0	$n < 40$: too few rows for eight bins of five; too few rows remain after deleting P12 (1 of 3 deletions), so the sign is untested there
halogenated | water as solute	18	3	1 (P13, 1.00)	+5.84	[+2.70, +9.43]	+3.39	0	$n < 40$: too few rows for eight bins of five; one source (P13), so there is no deletion to run and the sign was never put at risk
aryl ether | water as solute	13	2	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
halogenated | organic solute	12	4	1 (P3, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aromatic hydrocarbon | organic solute	8	8	1 (P11, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aromatic hydrocarbon | water as solute	6	1	1 (P13, 1.00)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
aliphatic hydrocarbon | organic solute	2	2	2 (P10, 0.86)	—	—	—	0	$n < 40$: too few rows for eight bins of five; no estimate at this n
The set as a whole								
whole set	477	185	15 (P13, 0.32)	+0.51	[−0.09, +2.16]	+0.95	3	passes here and fails at the pair unit (deployed convention), where deleting P13 takes +0.270 → −0.119

Table S12: **The map's error, bound and source curve**, stratum by stratum, at the same headline cell as Table S8—eight equal-count bins of $g(z^{\star})$, unbiased within-bin variance, row unit, deployed residual-only convention. These are the columns Tables S8–S11 do not carry, on the same 59 strata; margins, intervals and admissibility are there.

Stratum	n	MSE	$B^{\mathrm{up}}_{\mathrm{insuff}}$ [a]	P [b]	share [c]	leave-one-source-out [d]
Solvent class						
glycol ether	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water	111	0.477	0.215	≈ 0.8	5.8%	$[-0.19, +0.25]$ (3)
aprotic acceptor	57	0.089	0.140	< 0.2	0.6%	$[-0.41, -0.13]$ (4)
mono-alcohol	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
N–H protic	33	1.175	0.281	≈ 0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated	30	4.569	0.592	> 0.95	15.1%	$+5.84$ (one deletion)
aromatic hydrocarbon	14	4.297	—	—	6.6%	none possible
aryl ether	13	9.582	—	—	13.7%	none possible
aliphatic hydrocarbon	2	0.011	—	—	0.0%	none possible
Solvent class, coarse						
H-bond donor and acceptor (O)	328	1.651	0.562	≈ 0.9	59.7%	$[-0.42, +0.97]$ (8)
H-bond acceptor only	70	1.852	0.209	≈ 0.7	14.3%	$[-0.19, +2.94]$ (5)
H-bond donor (N)	33	1.175	0.281	≈ 0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated, non-H-bonding	30	4.569	0.592	> 0.95	15.1%	$+5.84$ (one deletion)
hydrocarbon, non-H-bonding	16	3.761	0.385	≈ 0.9	6.6%	none possible
Solvent family, fine						
glycol ether	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water	111	0.477	0.215	≈ 0.8	5.8%	$[-0.19, +0.25]$ (3)
mono-alcohol	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
ester	30	0.043	0.049	≈ 0.2	0.1%	none possible
amide N–H	25	0.591	0.414	≈ 0.8	1.6%	$+0.47$ (one deletion)
azine	23	0.096	0.072	≈ 0.5	0.2%	-0.00 (one deletion)
halogenated aliphatic	18	1.526	0.709	≈ 0.9	3.0%	none possible
aromatic hydrocarbon	14	4.297	—	—	6.6%	none possible
aryl ether	13	9.582	—	—	13.7%	none possible
halogenated aromatic	12	9.135	—	—	12.1%	none possible
amine N–H	8	2.999	—	—	2.6%	none possible
aliphatic hydrocarbon	2	0.011	—	—	0.0%	none possible
ketone/aldehyde	2	0.404	—	—	0.1%	none possible
nitrile	2	0.397	—	—	0.1%	none possible
Solute role						
organic solute	440	1.400	0.608	≈ 0.8	67.9%	$[-0.09, +0.41]$ (14)
water as solute	37	7.866	0.770	> 0.95	32.1%	none possible
Solute family						
alkane	129	2.315	0.467	> 0.95	32.9%	$[+1.39, +1.67]$ (7)
aromatic hydrocarbon	71	0.534	0.361	≈ 0.5	4.2%	$[-0.86, +0.08]$ (7)
ketone	63	0.628	0.577	≈ 0.5	4.4%	$[-0.14, -0.14]$ (2)
mono-alcohol	39	1.394	0.563	> 0.95	6.0%	$[+0.17, +1.37]$ (5)
water	37	7.866	0.770	> 0.95	32.1%	none possible
alkene	32	2.661	0.184	> 0.95	9.4%	$[+2.00, +2.68]$ (3)
aldehyde	29	0.091	0.144	≈ 0.4	0.3%	none possible
halogenated aliphatic	27	1.631	0.286	> 0.95	4.9%	$[+1.23, +1.71]$ (2)
amine N–H	12	3.362	—	—	4.5%	none possible
ether	12	0.469	—	—	0.6%	none possible
halogenated aromatic	8	0.619	—	—	0.5%	none possible
azine	6	0.135	—	—	0.1%	none possible
tertiary amine	4	0.278	—	—	0.1%	none possible
ester	3	0.013	—	—	0.0%	none possible
alkyne	2	0.350	—	—	0.1%	none possible
nitrile	2	0.124	—	—	0.0%	none possible
nitro	1	0.068	—	—	0.0%	none possible
Solvent class × solute role						
glycol ether | organic solute	182	2.252	0.108	> 0.95	45.2%	$[+1.60, +2.87]$ (3)
water | organic solute	111	0.477	0.215	≈ 0.8	5.8%	$[-0.19, +0.25]$ (3)
aprotic acceptor | organic solute	57	0.089	0.140	< 0.2	0.6%	$[-0.41, -0.13]$ (4)
mono-alcohol | organic solute	35	2.250	0.733	> 0.95	8.7%	$+1.95$ (one deletion)
N–H protic | organic solute	33	1.175	0.281	≈ 0.9	4.3%	$[-0.24, +0.87]$ (2)
halogenated | water as solute	18	6.294	0.228	> 0.95	12.5%	none possible
aryl ether | water as solute	13	9.582	—	—	13.7%	none possible
halogenated | organic solute	12	1.982	—	—	2.6%	none possible
aromatic hydrocarbon | organic solute	8	0.873	—	—	0.8%	none possible
aromatic hydrocarbon | water as solute	6	8.862	—	—	5.9%	none possible
aliphatic hydrocarbon | organic solute	2	0.011	—	—	0.0%	none possible
The set as a whole						
whole set	477	1.902	0.697	≈ 0.9	100.0%	$[+0.18, +0.87]$ (15)

[a] A dash is a stratum at which the fixed cell is undefined. Strata with $n \geq 40$ are *boundable*: eight bins hold at least five rows each. The others are reported anyway, and additionally carry the resample-adaptive rule $\max(3, \lfloor n/10 \rfloor)$ in the deposited table. On the mono-alcohols $B^{\mathrm{up}}_{\mathrm{insuff}} = 0.733$ exceeds that stratum's own $\mathrm{Var}(m) = 0.604$, which the artifact flags; it is the value that sets the $\times 6.8$ span of §3.2.1, and it occurs only on a stratum already not boundable. [b] Two-way solute × solvent cluster bootstrap, 3000 draws, quoted as a band by the rounding rule of §3. [c] That stratum's share of the set's squared error. Shares do not sum to 100% down the column: the granularities and the two axes overlap, which §3.2.1 measures. [d] Range of the margin over the deletions that can be run, with how many that is in parentheses; a deletion leaving fewer than 16 rows is not runnable, and a stratum with one source has none. Whether the sign survives—which is what admissibility turns on—is in Table S8.

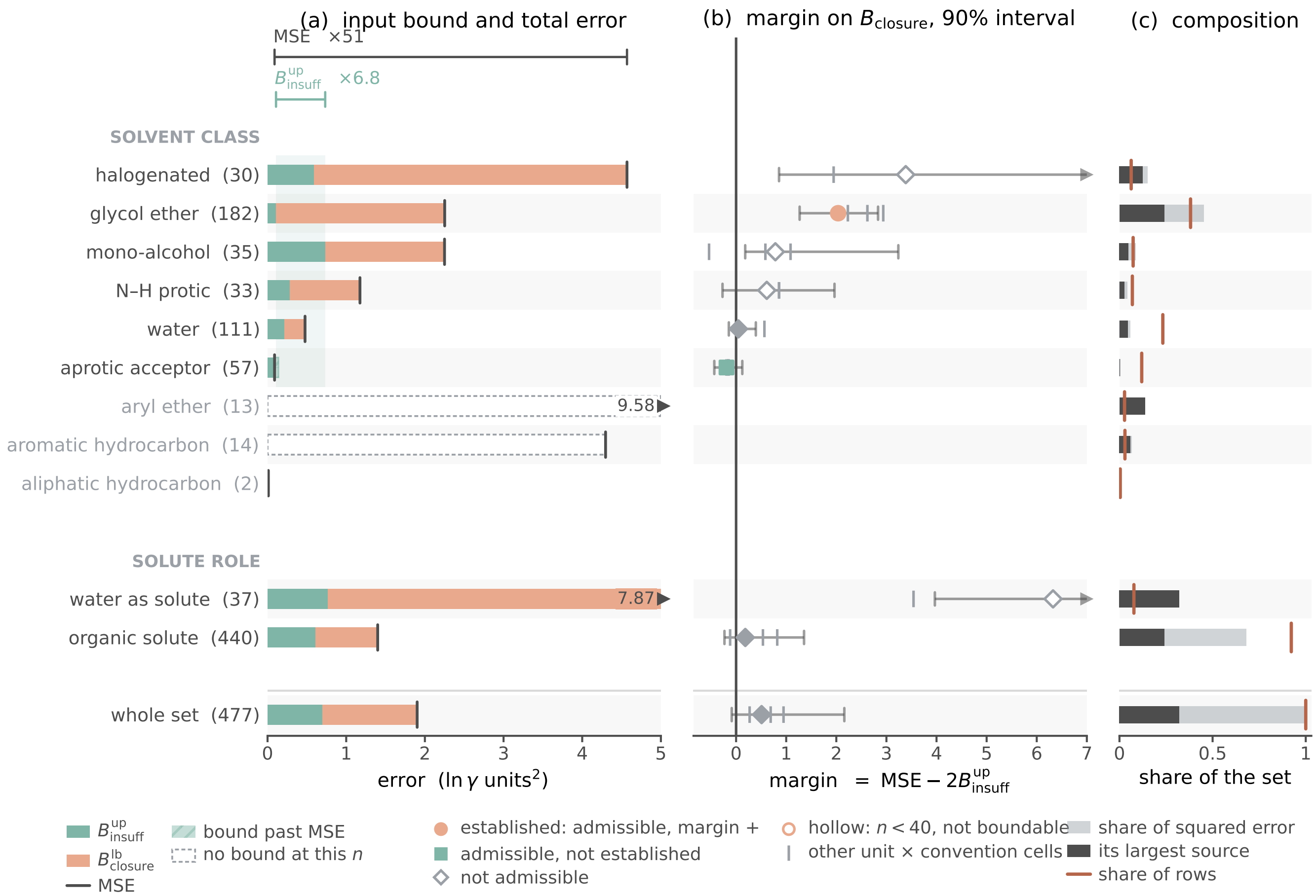

Figure S2: **Where the closure binds, and where it is not established**, by solvent class and by solute role, on the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Bars, points and shares are at the row unit under the deployed residual-only convention, at one estimator cell for every stratum—eight equal-count bins of $g(z^\star)$ with the Bessel within-bin variance. **(a)** each stratum's input-insufficiency bound and its total error; **(b)** the margin between them with its 90% two-way cluster-bootstrap interval, the other three unit × convention cells as grey ticks; **(c)** each stratum's share of the set's squared error, the part carried by its largest source publication drawn dark. This is the drawn form of the twelve cells at the reported granularity; the map itself, all 59 strata with their sources, intervals and admissibility, is Tables S8–S11 below. The marker in (b) is the admissibility verdict of §3.2.1, read off the same deposited columns the tables are built from. The one row set both positive and not contained in another is the glycol ethers, here at $P > 0.95$; no other row, the aggregate included, is established. *Caveat:* a hollow marker is a stratum below n=40—four of the drawn cells, with margins +0.78, +0.61, +3.39 and +6.33, none of them admissible—and three classes carry no bound at any n. The one established cell and the one admissible cell whose margin is not positive are both filled; panel (c) shows how much of each cell one publication carries. The bars on the two axes overlap: all 37 water-as-solute rows lie inside three solvent classes, which is why panel (c)'s shares do not sum down the column (Table S12, note c).

`10.1021/je020196d` is removed, so it is the glycol-ether sub-class and not the hydroxylic solvents as a class that carries the finding; the organic-solute subset flips to −0.09 under the same deletion; the mono-alcohols and the N–H protics are unboundable and each has a deletion that cannot be run; the aryl ethers are 13 rows from that same publication and carry no estimate at all; and the ester (30 rows, one publication, −0.06) and azine (23 rows, one deletion unverifiable) sub-families, which the acceptor-only reading would lean on, are inadmissible on the same grounds. Of the 59 strata only 15 are boundable at all, and 24 carry no estimate at any n this set supplies. The whole-set row is admissible at the headline cell and fails once the pair unit is required, where deleting `10.1021/acs.jced.7b00114` takes $+0.270$ to -0.119; the rule therefore does not by itself retire the aggregate at the headline cell, and its retirement rests on the composition argument of §3.2.1 rather than on this rule.

The two axes cross, and the map measures the crossing. 362 of the 1711 stratum pairs share rows, so cells drawn from the two axes are not independent evidence. All 37 water-as-solute rows and all six of its pairs sit inside three solvent classes and all come from one publication: 18 rows in the halogenated class (60% of it), 13 in the aryl ethers (all of it) and 6 in the aromatic hydrocarbons. Naming water-as-solute and the halogenated solvents as two findings would count those 18 rows twice, and adding the aryl ethers or the coarse acceptor-only class a further 13. The two positive admissible cells overlap the same way: alkane $\cap$ glycol-ether is 108 rows over 24 pairs and carries the whole effect ($+2.51$ on those rows), while the glycol ethers minus the alkanes remain admissible and the alkanes minus the glycol ethers do not. Those rows are counted once.

Concentration, inside the map as well as under it. Three solvent classes holding 47% of the rows carry 74% of the squared error, the 12% of rows that are acceptor-only aprotics carry 0.6%, and three of fifteen publications carry 76%. A map cell can be as source-carried as the aggregate was, so each is put through the same deletion (Tables S8–S11). Eleven multi-source cells have every deletion runnable in all four cells and six keep the sign throughout—the admissible ones. What singles the glycol ethers out among them is positivity and not being another admissible row set restated, not a deletion nobody else survives. On the halogenated solvents and the mono-alcohols only one of the two deletions can be run at all—in both, the one that cannot is the one removing the dominant source—and it leaves $+5.84$ and $+1.95$, so those two cells are unverified rather than shown to fail. At the level of the individual pair the broad set is stable, all 185 leave-one-pair-out margins falling in $[+0.42, +0.72]$, so what the aggregate is fragile to is the source and the stratum, which is what a composition effect looks like. The chromatographic-only cut that *inverts* the margin on the enlarged 298 K VT-2005-matched set (§3.6, limitation (i)) does not invert here under either convention.

The VT-2005-matched set, stratum by stratum and deletion by deletion. §3.2.2 states what this set is and that its aggregate does not stand; here is the arithmetic. The two worst-fit pairs each rest on a single record and carry 34% of the squared error, a concentration we do not read as established chemistry; the separation does not survive their deletion at the headline cell (margin -0.002), a margin of $+0.05$ being unable to absorb 34% of the squared error leaving the sample, and that deletion is a failure of the headline cell rather than a robustness pass. Four neighbouring cells do survive it, and the full composition audit is above. These are not 60 exchangeable observations, and the folds hinge on the same molecules. The stratification rule of §2.5 carries the composition half of §3.2.1 here and not the bound half: the 17 solvents span six of the nine classes—no water, no halogenated solvent, no glycol ether beyond ethylene glycol—and the largest reaches 24, so the set's composition is missing two of the three classes that carry most of the broad set's error, and of its 38 strata not one is admissible in every unit $\times$ convention cell. Within it, scored on UD profiles so that the two sets are commensurable, the ordering of the error is the broad set's—2.71 on the mono-alcohols, 3.00 on the N–H protics and 2.53 on the glycol ethers against 0.092 on the acceptor-only aprotics, that last cell again returning a negative margin, -0.05 at $P \approx 0.5$—but 57 of the 60 pairs are the broad set's own rows, so that agreement is a consistency check and not independent corroboration. What is this set's own is the granularity of its fragility: deleting the publication supplying 33 of the pairs takes the margin to -0.23. Scored against VT-2005 profiles under the deployed residual-only convention and the *unbiased* within-bin variance, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.713$ and the separation margin of $+0.05$; the maximum-likelihood variant biases the within-bin variance downward and so the margin upward to $+0.24$. That number stands with the broad set's aggregate, and on weaker evidence, four analyst choices setting it (§2.1.1).

Which defect the broad set points at, and why it states neither. §3.2.4 places the closure's failure on the VT-2005-matched set in the single-parameter H-bond kernel: 90% of that set's squared error falls on strong hydrogen-bond-donor solvents. On the broad set that reading does not reproduce, and neither does its replacement. The donor classes carry 64% of the squared error on 76% of the rows, which is no enrichment, while the halogenated and aryl-ether solvents—which the 2002 kernel has no dispersion term for at all—carry 29% on 9%, which is. But 31 of those 43 rows are `10.1021/acs.jced.7b00114`, and they are the same water-as-solute rows the map already declines to state; they carry 91% of that 29%. Put those two shares through the deletion the map applies to every cell and both invert: without that publication the halogenated and aryl-ether rows fall to 2.7% of the rows and 3.9% of the error, and the donor classes rise to 82% and 94%, which *is* the VT-2005-matched set's localisation. So one publication decides the reading in either direction, which is why the article states neither.

The constant-offset (Jensen) bound on both sets. §2.1.1 promises these here. On the broad IDAC set $\mathbb{E}[m] = 3.424$, $\mathbb{E}[g_{\text{res}}] = 2.998$ and $\mathbb{E}[g_{\text{full}}] = 2.579$, so the constant-offset bound is $B_{\text{closure}} \geq 0.182$ under the deployed residual-only convention and $B_{\text{closure}} \geq 0.715$ under the full one. Against the law-of-total-variance ceiling on B_{insuff} (0.697 residual-only, 0.482 full) the assumption-free bound therefore separates under the full convention and inverts under the deployed one, exactly as it does on the VT-2005-matched set (0.426 against 0.713) and by a wider margin—the deployed ratio is 0.26 here against 0.60 there. On this set it is in addition strictly slacker than the bound the decomposition already supplies, $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$ (0.182 against 1.205 residual-only, 0.715 against 1.429 full), so it adds nothing to the ordering under either convention; the deployed-convention separation rests on the law-of-total-variance bound alone, which is what §2.1.1 says.

The broad set's matching rule, and two bounds we decline to headline. Three details of §3.2.1 follow. *(a) The matching asymmetry.* The VT-2005-matched set is matched to VT-2005 by canonical SMILES, a rule we defend there on the ground that a profile imported across a tautomer or protonation variant is the wrong $z^\star$. The broad IDAC set is matched to UD by InChIKey with a first-block fallback, which is looser, and a wrong $z^\star$ inflates MSE and so B_{closure}—that is, it runs against the ordering we claim. The exposure is small and we measure it rather than argue it: of the 90 distinct molecules there, 86 resolve by full InChIKey and four only through the first block, touching 12 of the 477 rows. On the 465 rows matched exactly the verdict is unchanged—deployed convention MSE 1.930, $B_{\text{insuff}} \lesssim 0.70$, margin $+0.53$ at $P \approx 0.9$; full convention 1.926, 0.49, $+0.94$ at $P > 0.95$—so the looser rule is not carrying the result. *(b) The tighter of the two conventions' bounds.* B_{insuff} is the same population quantity under both conventions (step (d) of the proof of Lemma 1, §S2); what differs is the statistic the coarsening bins on, $g_{\text{full}}(z^\star, T)$ or $g_{\text{res}}(z^\star, T)$, each a deterministic function of the same argument and each therefore giving a separately valid upper bound. The smaller of the two is consequently also a valid upper bound in population, and charging it against the deployed error gives 1.902 against 2×0.48, a margin of $+0.94$ (on the VT-2005-matched set, 1.472 against 2×0.61, $+0.26$). We report this but do not headline it: a minimum over two finite-sample estimates is anticonservative even when each is valid in population, and the same-convention pairing is the conservative reading. *(c) The binning-free pair bound.* $\mathbb{E}[\mathrm{Var}(m \mid \mathrm{pair})] = 0.046$ is thirteen times tighter than the binning bound on the same 359 rows (0.61), and would put the margin *there* at +2.16 against the +1.02 the binning bound gives on those rows. It is not promoted because all 67 multi-row pairs draw their rows from a single publication, so it samples no between-laboratory spread (§3.2.1).

Whether the closure defect is correctable: within a solvent, not across one. Where B_{closure} dominates—on this set, the one row set of §3.2.1 that is established—the ceiling is in the map; that does not say the map is *fixably* wrong. We tested this with a small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel ($\Delta = B\,\mathrm{diag}(a)\,B^{\top}$ over the σ-grid, symmetric, zero-initialised), fit on the *reference* VT-2005 profiles and cross-validated five-fold over five fit seeds. The answer depends on the fold design, and only one design is free of leakage. With random folds over the 60 matched pairs, a rank-one residual (52 parameters) removes $46 \pm 5\%$ of the reference-input error out of sample (activity MSE $1.47 \to 0.79$). The same solvent, however, then appears in training and test, and the closure error is predominantly a between-solvent systematic. Under folds grouped so that no solvent is shared, the reduction is $-5 \pm 16\%$ at rank one and $+2 \pm 15\%$ at rank two. Each $\pm$ is a standard error over the five fit seeds on one fixed 60-pair set and prices optimiser noise only; the held-out spread across folds is an order of magnitude larger (sd 1.02 in MSE under the grouped design, 0.54 under random folds) and is not propagated into it. Stated as a bound rather than a null, the grouped design excludes a cross-solvent reduction above about 30%, and to that extent the apparent 46% is a within-solvent effect.

That fold spread also exceeds the gaps between ranks (≤ 0.06) by an order of magnitude, so these data do not resolve a rank ordering; and the 60 pairs form a single connected component under shared molecule identity, so a molecule-disjoint holdout does not exist at any fold design. B_{closure} is therefore a defect that a handful of kernel parameters can absorb *within* a solvent, and that we have not shown to be correctable across solvents—weaker than a general repair. The fitted correction also gains nothing over a free one: the parameter-free 2010 kernel reaches 0.765 on the same molecule pairs (§3.2.3) against the 52-parameter residual's random-fold 0.79. The residual's leakage-free number, however, is the grouped-fold one, a $-5 \pm 16\%$ reduction, i.e. no reduction at all, so the comparison that holds is a free $2002 \to 2010$ step against a fitted correction that does not transfer. The two also run on different profile sources and conventions (residual-only on VT-2005 here, full on the University of Delaware (UD) profiles there[S10]), so this is indicative rather than a head-to-head; but each stays above its own convention's insufficiency ceiling (0.62–0.71 residual-only, 0.53–0.61 full), so neither erases B_{closure}.

Where the residual error sits chemically. The composition audit behind §3.2: 90% of the VT-2005-matched set's squared error falls on the strong hydrogen-bond-donor solvents, 84% once the two doubted amine/methanol pairs are deleted, and the acceptor-only solvents are near-exact. That localisation is what places B_{closure} in COSMO-SAC-2002's single-parameter hydrogen-bonding kernel rather than spreading it over the profile. A counter-test on the same closure points the same way, and is the one §3.2 cites. The Staverman–Guggenheim combinatorial term is the physically *more* complete treatment and acts only where the two molecules differ in size; restoring it *increases* the VT-2005-matched set's reference-input error, MSE = 1.47 under the deployed residual-only convention against 1.80 with the term in place (Table S5), the gap between the two being carried by the size-asymmetric pairs. A term that adds physics and costs accuracy is what a misspecified map does and not what missing information about the inputs would do. The same signature appears on the solubility axis, where restoring the term makes the grounded learned-σ arm slightly worse (Table S2).

The closure-fidelity lever: the arithmetic behind §3.2.3. *(a′) The input-fixed kernel* *control.* The prescribed reduction of the typed Hsieh profile to a single 2002 profile is its sum $p = p_{\mathrm{nhb}} + p_{\mathrm{OH}} + p_{\mathrm{OT}}$ (Eq. 9 of Ref.[S1]), which holds database, cavity area, volume and averaging fixed and varies only the kernel. On the VT-2005-matched set that contrast is $2.047 \rightarrow 0.765$, two-way cluster bootstrap $P > 0.95$ with 90% margin $[+0.05, +2.90]$; the averaging step taken alone moves 2002 by $+0.29$ in the opposite direction and is not resolved ($P < 0.2$, $[-0.79, +0.11]$). That averaging term is why the control's 2002 endpoint is 2.047 and not the 1.757 of the *fair 2002* arm, and why the input-fixed step (1.28) is *larger* than the native-input one (0.99): the control is favourable to the conclusion it supports. The control cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction. What fixes the *fair 2002* arm at 1.757 rather than lower is an implementation detail: the NIST reference implementation, handed a typed three-profile database, silently reads the nonpolar channel only, an arm scoring 1.26 on the VT-2005-matched set (§S1 gives it unrounded), which is not the reference closure, so the arm is taken from our own NIST-validated layer instead. *(a) The* 2010, dispersion on *arm's coverage.* `cCOSMO` returns no dispersion value for the three bromine- or iodine-containing VT-2005-matched pairs—bromobenzene and iodobenzene in 1,2-ethanediol, and 2-bromopropane in pyridine—so that arm's VT-2005-matched run scores on n=57. On those common 57 the *2010, dispersion off* arm scores 0.767, so the 0.765-versus-0.785 ordering is not an artifact of the differing n. They are the *same* three pairs the broad set's construction excludes, its ladder also requiring a tabulated ϵ, so the dispersion-defined 57 and the 57 shared between the two sets are one set, not two. *(b) A rounding coincidence.* The 1.783 quoted for the shared 57 and the 1.783 quoted for the 334 chromatographic rows are distinct numbers that round alike (unrounded 1.78257 against 1.78245), the smaller set lying wholly inside the larger. *(c) The 2010 kernel on its own typed latent.* That latent is three 51-bin profiles per molecule, 153 in all, so the conditioning variable is 306-dimensional and the resolution ratios are $307/477 \approx 0.64$ per row and $307/185 \approx 1.66$ per distinct pair on the broad IDAC set, and $306/60 \approx 5.1$ on the VT-2005-matched set, where all rows sit at 298 K and T adds no dimension. Under the deployed residual-only convention the VT-2005-matched set gives 0.831 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 0.03$ and a margin of -0.77 at $P \approx 0.3$; the broad IDAC set gives 2.893 against $B_{\mathrm{insuff}} \lesssim 0.80$, hence $B_{\mathrm{closure}} \gtrsim 2.09$ and a margin of $+1.29$ at $P \approx 0.9$. Under the literature-standard full convention the same rows give 1.978 against $B_{\mathrm{insuff}} \lesssim 0.60$, margin $+0.78$ at $P \approx 0.8$, and the VT-2005-matched set 0.765 against 0.69, margin -0.61. The switch to the deployed convention raises the 2010 kernel's broad-set error from 1.978 to 2.893 and its floor bound from 0.60 to 0.80, so the wider margin comes from a worse closure and not from a tighter floor, and the reading does not move. On the broad set the binding clustering is the source-DOI bootstrap on 15 clusters, whose 90% interval $[-0.41, +4.53]$ still admits zero, as does the two-way one $[-0.24, +4.71]$; only the pair clustering is strictly positive, and it was so before the convention switch as well. *(d) The published comparison.* The comprehensive assessment[S10] covers 29,173 γ^{∞} points for the 2010 and dsp kernels only. Its largest documented infinite-dilution step—adding the whole dispersion term—yields a ∼10-point mean-absolute-deviation gain (95% → 86%) with a *reversal* on aqueous systems (203% vs. 194%); 2010's gains are otherwise concentrated in liquid–liquid equilibria and in the temperature dependence of the electrostatic term, where an infinite-dilution design has little leverage. *The scope of the search behind the null of §3.2.3*: the papers introducing each kernel,[S2,S3,S11] the benchmark implementation,[S1] this comprehensive assessment[S10] and the works citing them. It was not exhaustive, which is why the main text reads the result as an absence of evidence rather than as evidence of absence. *(e) The floor checks.* Each is same-database, same-latent and same-convention, stated under the deployed residual-only rule with the full-convention reading beside it. *(1)* Broad IDAC set: the fair 2002 error 1.902 against $2B_{\text{insuff}} \lesssim 1.39$ on the *same* UD profiles, margin +0.51 at $P \approx 0.9$ (1.911 against 0.96, +0.95, $P > 0.95$ full). *(2)* VT-2005-matched, VT-2005 against VT-2005: 1.472 against $2B_{\text{insuff}} \lesssim 1.427$, so $B_{\text{closure}} \gtrsim 0.759$ against $B_{\text{insuff}} \lesssim 0.713$ and a margin of +0.05 (1.80 against 1.22, +0.58 full; Table S5). The 1.757 of (a′) is those same 60 pairs scored on *UD* profiles, which is the whole difference between 1.80 and 1.757. Charged against a UD-native floor recomputed for those pairs, the deployed convention gives 1.436 against $B_{\text{insuff}} \lesssim 0.71$, a margin of +0.01 at $P \approx 0.8$, against 1.757 versus a UD-native $B_{\text{insuff}} \lesssim 0.54$ and a margin of +0.69 full—so on UD profiles the VT-2005-matched set's own floor check is, under the rule we deploy, at zero.

Robustness of the closure/insufficiency separation. We stress-test the deployed-convention verdict $B_{\text{closure}} > B_{\text{insuff}}$ (§3.2)—equivalently $B_{\text{insuff}} < \text{MSE}/2 = 0.74$ at $\text{MSE}_{\text{res}} = 1.47$—four ways on the n=60 set.

(i) Estimator sensitivity. The four random-fold B_{insuff} upper bounds of Table S7 (the LOTV one is also the headline row of Table S5)—0.40 (k-NN), 0.71 (binning/LOTV at the headline unbiased variance; 0.62 at the maximum-likelihood one), 0.62 (random forest), 0.62 (ridge out-of-fold)—all fall below the 0.74 threshold, the binning bound only just. The forest is the one estimator our two VT-2005-matched audit artifacts disagree on: the standalone decomposition run's own forest gives 0.565 on these rows against the audit's 0.621, a spread of the size a forest's own randomisation produces. The audit value is the one Table S7 carries, because the audit artifact is what the inclusion rule names; both are below the threshold and neither is the headline. The k-NN value is the smallest and least conservative input to $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$, so we take the leakage-immune LOTV bound as the headline instead. The LOTV bound bins $g(z^\star)$ and is therefore finite-sample downward-biased when points per bin are few, which would *inflate* B_{closure} favourably. The two non-binning bounds (RF 0.62, ridge 0.62) are immune to that particular per-bin sparsity, but they do not provide independent corroboration: under random folds every held-out row shares its solvent or its solute with a training row, so both can borrow strength from a near-twin, anti-conservatively and in the same direction as the binning (§3.2). Blocking folds on both molecules leaves no informative estimator on this set. The broad IDAC set behaves the same way once its own leak—a held-out row's molecule pair recurring at a neighbouring temperature—is blocked: the random forest goes from 0.15 to 0.88 and the ridge from 0.80 to 1.47, past its threshold. The headline should therefore be read as resting on the leakage-immune LOTV bound alone, on either set, not on agreement between three estimators. The sole non-certifying configuration is a deliberately aggressive ridge, which lifts B_{insuff} to 0.88 (then $B_{\text{closure}} \geq 0.59 < 0.88$): a sensitivity edge, not a refutation.

(0) The VT-2005-matched set's design. The 60 pairs span 41 solutes and 17 solvents; two formamide pairs are excluded for a canonical-SMILES tautomer mismatch. Methanol occupies 25 of the 60 rows (12 as solute, 13 as solvent) and pyridine is the largest solvent stratum (20 rows); $\text{Var}(m) = 2.47$ across the set. The two worst-fit pairs, 1-hexanamine and cyclohexylamine in methanol, carry 34% of the squared error; both rest on a single record, and 1-hexanamine sits far off its own homologous series ($\ln \gamma_2^\infty$=2.62 against −0.78 for 1-butanamine in the same solvent), which is why we do not read that concentration as established chemistry. Deleting both leaves MSE 1.01; charged against this the headline cell of the deployed convention—eight equal-count bins under the *unbiased* within-bin variance—gives $B^{\text{up}}_{\text{insuff}}$ 0.5056 and margin −0.0015, so the separation does *not* survive the deletion where the paper headlines it. The four neighbouring cells do: 0.4370 (margin +0.14) at eight bins under the maximum-likelihood variance, 0.4717 (+0.07) at six bins unbiased, 0.4241 (+0.16) at six bins maximum-likelihood, and, under the full convention on the same 58 rows, MSE 1.20 against 0.3309 (+0.53) at eight bins unbiased. The reading we take from the cell that fails is

that the VT-2005-matched set's +0.05 headline margin has no room to absorb the loss of 34% of the squared error, which is the conclusion the leverage folds below reach by a different route.

(i bis) Leverage folds on the VT-2005-matched set. The separation survives all 17 leave-one-solvent and all 60 leave-one-pair folds and deletion of the eight highest-error pairs, but only 40 of 41 leave-one-solute folds (six equal-count bins): the exception drops methanol, the maximum-leverage solute (12 pairs), and loses the separation (-0.26), as does dropping methanol as solute with pyridine as solvent (n=28; -0.36). Deleting methanol from both roles (n=35) holds only marginally—the margin falls from 0.35 to 0.11 at six bins and from 0.24 to 0.003 at eight, and inverts under the Bessel-corrected variance (-0.03, -0.21). Those fold counts and the bootstraps use the maximum-likelihood variance at six bins. Repeated at the headline cell—eight bins, unbiased variance—the leave-one-solvent count is unchanged at 17/17 (minimum margin +0.03), the leave-one-pair count falls to 56/60 (minimum -0.03) and the leave-one-solute count to 36/41 (minimum -0.22); deletion of the eight highest-error pairs still holds there (+0.04).

(ii) Leave-one-solvent-out. Dropping each of the 17 solvents in turn, the VT-2005-matched set's aggregate margin stays positive in all 17 folds under the LOTV bound—a fold curve on a quantity §3.2.1 retires, so it records what that aggregate did and not a robustness pass. Under the random-forest estimator it holds in only 2/17, because an RF over-fits conditional variance in 102 dimensions at n=60 (its out-of-fold MSE approaches $\mathrm{Var}(m)$), so its leave-one-out estimate is unstable—an estimator artifact, not a sign flip.

(iii) Label noise. Injecting zero-mean Gaussian noise $\sigma_\eta \in \{0.2, \ldots, 0.8\}$ into m leaves the B_{closure} lower bound stable ($0.84 \to 0.70$), as expected since $B_{\text{closure}} = \text{MSE} - \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ is invariant to zero-mean noise.

(iv) Clustered-bootstrap interval. A solvent-clustered bootstrap (3000 resamples of the 17 solvents, LOTV bound) makes the confidence explicit: the B_{closure} lower bound has median 0.93 (CI_{90} [0.40, 1.62]), and the margin $B_{\text{closure}} - B_{\text{insuff}}$ has median 0.36 (CI_{90} [-0.22, +0.97]) with $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.89$ (single-axis; the conservative two-way solute×solvent bootstrap, which also charges for repeated solutes, gives ≈ 0.78, the value reported in the main text). With $G = 17$ solvent and 41 solute clusters this small-G regime has optimistic interval coverage, and the finite-cluster correction acts in the direction of less confidence and non-certification. We therefore treat the P values as a coarse band, not two-decimal quantities, and the main text quotes them as bands (rounded to one decimal, with > 0.95 and < 0.2 closing the ends) throughout; the two-decimal values are in the deposited artifacts and in the tables here, where the estimator that produced each is named. The separation on the VT-2005-matched set does not clear the 90% level under either clustering, which is why the article treats it as supporting rather than decisive. On the broad IDAC set, with 78 solute and 36 solvent clusters and a third clustering over 15 source publications, the bands mean something; but they do not all clear. Under the deployed residual-only convention the margin bootstraps to $P > 0.95$ over the 185 molecule pairs (90% margin [+0.23, +1.16]) and only $\approx$0.9 two-way over the 78 × 36 solute/solvent clusters ([-0.08, +2.16]) and over the 15 sources ([-0.07, +1.74]), with two of the three 90% intervals extending just below zero, so that set is not resolved either. Read as a ratio, the broad set's +0.51 is 1.1 times its own 90% bootstrap half-width under pair clustering and 0.46 times it under the two-way one, which is the sense in which §3.2 calls that margin unresolved. It is only under the full convention that all three clear ($P > 0.95$), in the same order: [+0.71, +1.27] over pairs, [+0.48, +1.85] two-way, [+0.50, +1.57] over sources.

(v) A label systematic tracking $z^\star$, and what the VT-2005-matched set's fold designs leave. Zero-mean noise cannot manufacture the separation: it raises MSE and $B^{\text{up}}_{\text{insuff}}$ alike, leaves the B_{closure} lower bound unchanged, and shrinks the margin by σ_η^2, so it can understate the separation but not create it. What the margin does bound is a *systematic* label error correlated with $z^\star$, which enters B_{closure} rather than B_{insuff}: on

the VT-2005-matched set it would close the separation at root-mean-square 0.49 $\ln\gamma$ under the maximum-likelihood binning but at only ≈0.21 under the Bessel-corrected one. The matched IDAC is essentially single-method (158/165 gas chromatography), which makes a within-method systematic of 0.5 implausible; one of 0.21 is not, so on the corrected binning that threat is *not* excluded. The VT-2005-matched set's out-of-fold estimators cannot police it either: under random folds every held-out row shares its solvent (47/60 = 78% of rows) or its solute (27/60 = 45%) with a training row, and under folds blocked on both molecules three of the four estimates (3.65, 1.83, and the divergent ridge) exceed the total MSE = 1.47, beyond which an upper bound on B_{insuff} constrains B_{closure} vacuously.

In-regime (γ-stratified) separation. The n=60 measurement is matched at low γ, while the paradox spans all γ. Stratifying the n=60 set by $\ln\gamma_2^\infty$ tests whether closure-dominance is a regime artifact. It is not: the ordering $B_{\text{closure}} > B_{\text{insuff}}$ (leakage-immune LOTV binning bound) holds in every γ stratum and *strengthens* toward higher γ (Table S13)—in the highest stratum ($\overline{\ln\gamma_2^\infty} = 2.94$, approaching the non-ideal regime) $B_{\text{closure}} \geq 2.68$ exceeds $B_{\text{insuff}} \leq 0.67$ several-fold. The high-γ strata are small (n = 17–30 in 102 dimensions): the law-of-total-variance bound can be pushed toward zero by too-few-points-per-bin, which mechanically widens the margin exactly where data is thinnest, so it is the several-fold separations rather than the precise values that carry the qualitative reading. The bound here bins each stratum into three equal-count bins (against eight in the main text) and uses the maximum-likelihood within-bin variance, as in the headline row of Table S5; recomputed with the unbiased (Bessel) one it rises to 0.08/0.95/0.81 across the three strata, so each still separates, but the middle stratum only marginally (0.95 against its own MSE/2 = 0.96). The $m >$ med. stratum separates through its upper (high-γ) half, and the *isolated* moderate band (med.$< m \leq 1.5$, n=13) does *not* separate. This is not a counter-example. The total error there is only ≈0.06, so $B_{\text{closure}} \leq \text{MSE} \approx 0.06$ assumption-free (since $B_{\text{insuff}} \geq 0$), and there is nothing to attribute—near ideality the target is small and the split is uninformative. We read no closure-quality claim from it in either direction. This supports—but does not prove—the conjecture that the same closure misspecification drives the all-γ paradox.

Table S13: Closure/insufficiency verdict across $\ln\gamma_2^\infty$ strata of the matched set (deployed residual-only convention; $B^{\text{up}}_{\text{insuff}}$ is the leakage-immune LOTV binning bound throughout, at the maximum-likelihood within-bin variance).

Stratum	n	$\overline{\ln\gamma_2^\infty}$	$B_{\text{insuff}} \leq$	$B_{\text{closure}} \geq$
lower γ ($m \leq$ med.)	30	−0.40	0.07	0.95
higher γ ($m >$ med.)	30	+1.90	0.85	1.07
high γ ($m > 1.5$)	17	+2.94	0.67	2.68

The closure/insufficiency decomposition (full/res conventions; law-of-total-variance (LOTV), random-forest (RF), ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum) is tabulated in §3.2. Figure S3 shows the shape of that error on the VT-2005-matched set: the reference-σ closure's parity plot against the measured $\ln\gamma_2^\infty$.

Four qualifications on the decomposition (full). Four qualifications bear on the decomposition of §3.2. None of them inverts the sign of the VT-2005-matched set's aggregate margin; each sharpens its scope. That aggregate is retired on other grounds (§3.2.1–§3.2.2), so these are recorded as what the four choices do and not as robustness passes for a live verdict.

1. The deployed combinatorial default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.71$; it separates only under the full convention, $B_{\text{closure}} \geq 0.72$ against $B_{\text{insuff}} \leq 0.61$), so that assumption-free bound does not separate under deployment; Table S5 carries that row for both sets, and the full-convention margin (the sharpened MSE−

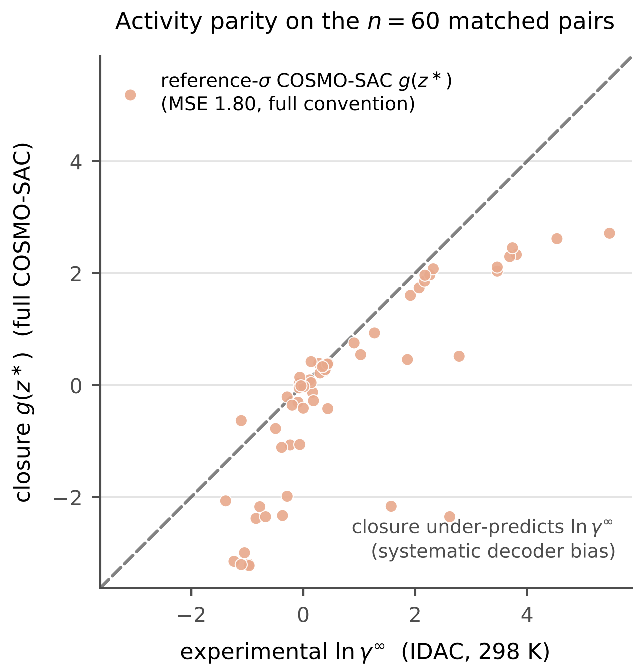

Figure S3: Activity-coefficient parity on the n=60 matched pairs. The reference-σ COSMO-SAC closure $g(z^\star)$ (salmon) falls below the diagonal at larger $\ln \gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention).

B_{insuff} bound, 1.19 vs 0.61) is a full-COSMO-SAC-only result. What separates under the deployed convention is the leakage-immune bound $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{LOTV}} \approx 0.76 > 0.71$, which carries the headline claim, together with the combinatorial argument above. Every B_{insuff} ceiling quoted in this item is the unbiased within-bin variance, the headline setting of Table S5; the maximum-likelihood pair (0.85 against 0.62) is the demoted variant and is not quoted here.

2. The closure error is predominantly a *between-solvent* systematic bias ($\sim$64% of total error); within the largest single-solvent stratum (pyridine, n=20) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of $z^\star$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).

3. Confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 vs the random-forest upper bound, 0.89 vs the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The leakage-immune margin (0.76 vs 0.71) is a point estimate. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. The 60 matchable pairs are the low-to-moderate γ regime (matched mean $\ln \gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln \gamma_2^\infty$ at 298 K, so the numeric ceiling does not necessarily transfer unchanged to the finite-composition solubility regime.

S4 External baselines and ablations

We evaluated the direct state-of-the-art model (FastSolv[S12]) and the physics-informed state-of-the-art model (SolProp[S13]) on a by-solute test split (`test_solute.csv`, $n \approx$ 10–12k). The two arms are not obtained the same way, and the difference matters below: FastSolv is retrained on our corpus, whereas SolProp is the released pretrained thermodynamic cycle—the arm that instantiates the physics—mapped onto our target by a three-parameter linear recalibration fitted on the training rows, not by retraining. That cycle is run at 298 K on every row, so the temperature of a measurement reaches the SolProp arm only through the calibrator's temperature term. In $\ln x_2$ they reach MAE 1.94 (FastSolv, R^2 0.23) and 2.34 (SolProp, calibrated, R^2 −0.18); in $\log_{10} S$, 0.87 and 1.04. These are on a *by-solute* split that is *not comparable* to our scaffold numbers (Table S2): they bound the task's difficulty, not a ranking of the models against one another, so this does *not* establish that our scaffold-split DirectGNN (1.70) outperforms them. A like-for-like scaffold evaluation of the external models was not run, the deposited external runs being by-solute and pair-random only. The internal ordering—the physics-informed SolProp trailing the direct FastSolv, the same ordering we find between our physics closures and DirectGNN—is therefore a comparison of the *cycle* against a direct model. It reverses when SolProp's encoder is instead retrained directly on $\ln x_2$ with the cycle discarded: that arm reaches MAE 1.63 (R^2 0.39) in $\ln x_2$ and 0.76 in $\log_{10} S$ on the n=10,287 rows FastSolv is scored on, against FastSolv's 1.94 and 0.87. We quote the cycle as the physics-informed arm because only it assembles the separately learned components; the retrained arm measures what discarding the closure achieves on this corpus, not a better physics-informed model. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §3.6. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table S2.

S5 Supplementary mechanism results

The activity map's Fisher geometry, and why σ is under-determined. At infinite dilution the aggregate activity Fisher $J^\top J$ ($J = \partial \ln \gamma_2^\infty / \partial \sigma$) is rank-one (participation ratio 1.0 by autograd; finite differences decorrelate the redundant flat-profile coordinates and inflate it, §S6.6, so we read PR against a permuted control and not absolutely). The per-molecule Fisher that governs the drift has participation ratio $\approx$1.4 of 51 even at full solvent coverage. The participation ratio is a concentration statistic and not an exact count. It is measured at infinite dilution, so its transfer to the finite-composition regime that generates the drift is not established, the same scope gap that qualifies B_{closure} (§3.6). The external reference is itself *computed* (VT-2005 quantum chemistry) rather than experimental, so none of this establishes that recovery of the physical profile is impossible in principle.

Why the top-two share is read against a phase-randomised null, and how unstable the magnitudes are. A top-two explained-variance share is not interpretable against an isotropic null, because smoothness alone concentrates a spectrum: smoothing isotropic noise reaches a top-two share of 0.25 to 0.83 as the correlation length grows from one bin to eight, spanning the observed value without any between-molecule alignment at all. The *phase-randomised* null of §3.3 preserves each deviation's amplitude spectrum exactly—and so its smoothness—and destroys only the alignment across molecules that a shared compensating direction asserts; the observed 0.399 against its 0.221 ± 0.011 is therefore a statement about alignment and not about smoothness. The wrong-molecule null (0.772 ± 0.023), which the observed share does *not* clear, is a difference between two points on the reference manifold and inherits that manifold's own concentration, so it demands more alignment than a shared direction implies. That is the argument for preferring the phase-randomised one, and §3.3 states that if it is not accepted the spectrum is uninformative rather than supporting. In each case the ± is the null's spread over resamples, not an uncertainty on the observed value. The three magnitudes §3.3 reports but does not carry move in both directions across analysis configurations: a shortened run gives only ∼16% departure and a transfer of 1.7×, below what a constant offset alone reaches on the one run where that control can be computed, while uncontrolled two-model comparisons over-state both the departure (∼82%) and the transfer (16–26×). With n=3 the ± on the three-seed figures is indicative and not a precision. Nor does mean-centred low-rank structure exclude a shrinkage of each profile toward the corpus-mean shape: the reference profiles themselves sit at 0.77 mean-centred, so any drift linear in the profile inherits that concentration.

The drift spectrum, and the null it lacks. Figure S4 is the spectrum of the drift whose

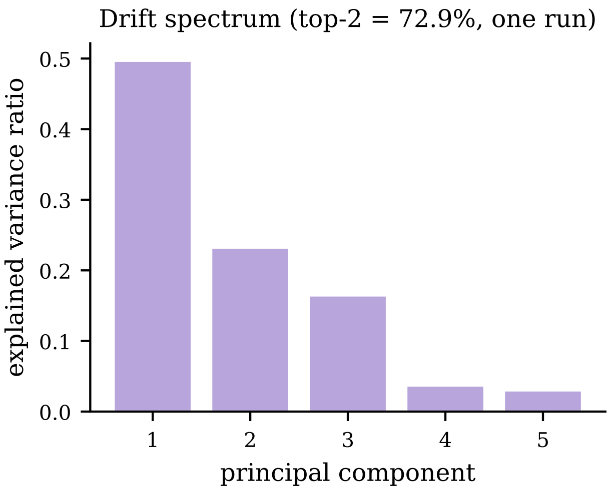

Figure S4: Explained-variance spectrum of the fine-tuning drift on the 44 matched molecules (companion run). *Caveat:* this share carries no matched null (text).

mean and dominant mode are Fig. 4. Its top two modes carry 72.9% in the companion run plotted there and $71.8 \pm 1.3\%$ over the three seeds (quoted as $72 \pm 1\%$ in §3.3): the same statistic on different runs. The companion run's spectrum is `results/compensation/isolation_gpu.json` and the three-seed aggregate `results/sur/surrogate_seeds/surrogate_seeds.json`, whose `top2_evr` has mean 0.71819 and standard deviation 0.01265 over seeds 0–2; both this figure and Fig. 4 are drawn by `scripts/analysis/make_surrogate_figure.py`. That share has no matched null, so the panel is titled by what it plots and not by an inference. The phase-randomised and wrong-molecule nulls of §3.3 are run on the $\hat{\sigma}$-versus-reference deviation instead, a different quantity; the isolation drift has not been re-tested that way, because the two locally available checkpoints share a σ head and so have identically zero drift between them.

The grounded warm-up checkpoint's 0.73, and the power of the two-MSE check. The direct cancellation check of §3.3 has almost no power in the direction that matters. On the VT-2005-matched set, 18.75 against $\mathrm{Var}(m) = 2.47$ is $7.6\times$ worse than predicting the mean, the signature of a broken profile rather than of a partly compensating one, so the comparison separates "no cancellation" from "large cancellation" and not from "some cancellation". The same comparison run on a grounded warm-up checkpoint, whose profile sits 28% from the reference, scores 0.73 where the end-to-end SLE-trained checkpoint scores 18.75. That is a statement about the closure's sensitivity to profile roughness and about an in-sample fit to $z^\star$ on molecules inside the grounding pool, not evidence that the encoder carries information the reference lacks, and we do not read it as a both-channel reproduction of the paradox.

The closure's own-axis check on polarity. The second check of §3.3: on the n=60 VT-2005-matched set the reference-input closure *under*-predicts, its mean output being +0.10 under the residual-only convention and −0.10 under the full one against a mean target $m = +0.75$. Sweeping its hydrogen-bond coefficient at zero, one and two times the standard value gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs: raising the coefficient makes them worse and zeroing it makes them better, so a *more* polar profile moves g further from m, which is the direction opposite to a compensating drift.

The reference ladder. A relative deviation carries no absolute scale, which is what the ladder supplies. Its rungs are built from the reference profiles the grounded arm was supervised on, so it calibrates that deviation rather than competing with it. The rungs calibrating the 51% departure of §3.3, all on the same 44 molecules and the same relative-L_2 metric: a leave-one-out corpus-mean profile (one fixed shape reused for every molecule, carrying no molecular information) 0.493; each molecule assigned another molecule's reference profile, 0.651–0.684 over five derangements; a uniform profile 0.889. Against these, the grounded base sits at 0.364 and the fine-tuned latent at 0.507. The grounded base is the σ-supervised recovery §3.3 reports as reproducing the reference profile far more closely: it is $36 \pm 1\%$ over the three seeds in the norm ratio $\|\hat{\sigma}_{\text{grounded}} - \sigma\|/\|\sigma\|$, and it is an *in-sample* fit—all 44 probe molecules lie in the σ-grounding pool and the warm-up early-stops on it—so no out-of-sample recovery estimate exists on this set.

The distortion operator D. The transfer test of §3.3 fits a dense affine map on the 51-bin grid: a 51×51 linear part plus a 51-bin offset, 2652 parameters, each output bin a ridge regression of 52 coefficients (ℓ_2 penalty 10^{-3}) on 22 of the 44 molecules. The constant-offset control replaces D by a single molecule-independent shift of the whole profile, fitted on the same 22 molecules and applied unchanged to the held-out 22. It can be computed on one companion run only—the two-model comparison whose two checkpoints are both still on disk—and there it is 2.1× better than the identity null on both of that run's arms. That run is the uncontrolled one, whose own transfer of 16–26× §3.3 reports as over-stated, so its residual factor is not read across. The three seed checkpoints behind the $4.1 \pm 0.5\times$ were not retained, and the remaining companion run records only an ephemeral checkpoint path, so no offset control exists for either and the 4.1× is a ratio against the identity null with no constant-offset control of its own. The homologue saturation that makes "held out" largely within-cluster interpolation is six n-alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44; the median held-out molecule sits 0.048 in L_2 from its nearest training-half neighbour, about 16% of its own profile norm.

Non-identifiability, empirically (the corrupted twin). Training a twin model on data whose crystal labels are corrupted (every T_m shifted by +273 K) leaves the *solubility* fit essentially unchanged—$\ln x_2$ MAE 1.835 versus 1.812 on the corrected data—while the recovered crystal grounding differs by hundreds of kelvin (T_m MAE 258 K versus 45 K). Two very different crystal/activity splits produce the same solubility, which is Lemma S1 made concrete.

Crystal grounding on physical labels. When the crystal head is supervised on measured melting points it predicts T_m to 45 K MAE, about 4× better than a Joback group-contribution reference—so the flat downstream result of §S6.3 is a dose/identifiability effect, not an inability to learn crystal properties.

The $\Delta C_p = 0$ approximation. The ideal term omits the constant-heat- capacity correction; a group-contribution audit finds a nonzero ΔC_p for 1441 of 1525 solutes (108,287 rows), so this is a bounded model-form approximation. It cannot produce either headline result. The closure/insufficiency decomposition is measured on the activity target ($\ln \gamma_2^\infty$, at 298 K on the VT-2005-matched set and at each row's own temperature over 273–403 K on the broad IDAC set), where ΔC_p does not enter at all on either set: the ideal term is not evaluated in the decomposition, so no temperature range of it is at issue. The grounding paradox is an arm-to-arm difference (learned-σ vs. oracle-σ) whose two arms share the *identical* crystal term, so any ΔC_p error cancels exactly in the comparison. It is thus orthogonal to the activity closure, not a source of the paradox.

Fusion-supervision scarcity. Direct ΔH_{fus} labels are extremely sparse: 1279 training rows across 31 solutes, and essentially none in validation or test. This is the quantitative basis of the identifiability argument—there is almost no signal with which to pin the crystal term from data.

The NRTL branch trains (a trainability check). The asymptotic physics penalty is not an under-training artifact of the activity branch. In the headline NRTL arm the head produces solvent-varying interaction parameters—τ_{12} over the n=8103 test pairs has mean 0.97 and std 2.5 (range $[-2.6, 22]$), far from collapsed—yet the arm still trails the closure-free DirectGNN (1.80 vs 1.70; Table S2). A trained, non-degenerate activity branch that the control still outperforms is evidence that the penalty is structural, not a failure to train the branch. (The core closure-misspecification result—the σ-oracle and the decomposition—is in any case training-*independent*: it feeds the reference VT-2005 profiles through the fixed closure with no encoder optimisation, so it cannot be a training pathology.)

Structured temperature extrapolation. Withholding *temperature* rather than structure (train at $T \leq 309.75\,\mathrm{K}$, test at higher T; $n = 3343$), the affine-in-$1/T$ hypothesis classes extrapolate far better than an unstructured descriptor regressor (Table S14); the van't Hoff class also predicts the *direction* of the high-T shift correctly 99% of the time versus 40% for the random forest. This is a property of the structured hypothesis class—the trained neural models are weaker than the van't Hoff baseline—not an accuracy claim for our model.

Table S14: Same-pair high-temperature extrapolation ($n = 3343$, train $T \leq 309.75\,\mathrm{K}$). "dir." is the fraction of pairs whose high-T shift direction is predicted correctly.

Model	MAE	R^2	dir.
per-pair van't Hoff ($1/T$)	0.37	0.89	0.99
per-pair linear in T	0.41	0.85	0.99
per-pair last-low-T hold	1.20	0.69	0.01
RF (Morgan $+\,T$)	1.29	0.66	0.40
per-pair mean	1.46	0.56	0.01

S6 Supplementary diagnostics and broad negatives

S6.1 Supporting diagnostics

The interpretive claims of the preceding sections are based on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that are unfavourable to the physics-informed hypothesis.

Aleatoric noise floor. The derivation, the two aggregating functionals and their values are in the article at §2.2. One property of the per-group distribution is not: it is strongly right-skewed, which is why the mean and the median differ by a factor of five. On the groups backed by two sources the 90th percentile of the per-group disagreement is 0.42 $\ln x_2$ units and the 99th is 1.41; on those backed by three, 0.83 and 3.55. The floor's mean is therefore set by a minority of badly disagreeing systems rather than by a broad band, and on either reading it lies far below the scaffold-split error.

The encoder is transfer-limited, not expressivity-limited. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; what degrades out of distribution is the generalisation of that expressivity across scaffolds. This evidence is suggestive rather than decisive: the sharp per-descriptor spread (FractionCSP3 0.96 vs HeavyAtomCount 0.23) means the train–test gap partly reflects covariate shift in the descriptor *targets* across scaffolds rather than an encoder property alone, and the probe measures descriptor-representability, not task-relevant expressivity directly. It therefore *suggests*—rather than establishes—that a new graph encoder is unlikely to be the main lever for out-of-distribution accuracy, with pretraining and coverage the more likely ones.

Temperature extrapolation. A structured van't Hoff activity class (log-solubility affine in $1/T$) encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this is apparent when the split withholds temperature rather than structure. Training on measurements at $T \leq 309.75$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.37 against

Table S15: Solvent-ranking quality of the physics path on the scaffold test ($n = 826$ solute groups).

Metric	Value
Spearman ρ	0.511
Top-1 accuracy	0.477
Kendall τ	0.436
nDCG@3	0.732

1.29 for a random forest over Morgan fingerprints plus temperature (in $\ln x_2$ units), consistent with the temperature-aware $\ln \gamma_2^\infty$ modelling of.[S14] This is a property of the structured hypothesis class, not a performance claim for the trained model: the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Ranking and calibration. Absolute error and decision quality diverge. Despite its high MAE, the physics path orders candidate solvents usefully (Table S15; nDCG@3 $\approx$ 0.73 across $n = 826$ solute groups). Uncertainty, by contrast, is badly miscalibrated in the native model—a nominal-90% interval covers only PICP@90 = 0.13 of held-out points—but split-conformal wrapping repairs it to 0.92 empirical coverage at the same nominal level. The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

S6.2 Identifiability and compensation

Lemma S1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and compute the conditioning cond(I) and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes.

Table S16: Fisher audit of the SLE design at $\Delta T = 40\,\text{K}$ (synthetic, closure-independent).

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ T_m + ΔH_{fus}	1.6×10^3	1,689

Every entry below is conditional on a design that must be stated, because a Cramér–Rao bound in J/mol scales directly with the assumed measurement noise. The operating point is $\Delta H_{\text{fus}} = 25{,}000\,\text{J/mol}$, $T_m = 400\,\text{K}$, and an activity term $\ln \gamma_2^\infty(T) = a + b\,(T_{\text{ref}}/T - 1)$ with $a = 1$, $b = 200\,\text{K}$ and $T_{\text{ref}} = 298.15\,\text{K}$; x_2 is not a free input but the SLE solution at each temperature, running from 3×10^{-8} to 0.15 across the window. The design is $n = 9$ equally spaced temperatures spanning $\Delta T = 40\,\text{K}$ from 280 K, one measurement each. The noise levels are $\sigma_{\ln x_2} = 0.3$ on each log-solubility observation, $\sigma_{T_m} = 10\,\text{K}$ on a melting-point label and $\sigma_{\Delta H_{\text{fus}}} = 2{,}000\,\text{J/mol}$ on an enthalpy label. Parameters are non-dimensionalised by the scales used in the model's own configuration (5,000 J/mol, 50 K, 1, 300 K), so the condition number reports rank structure rather than unit choice. Table S16 reports the result at that window. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label reduces the conditioning by nearly two orders of magnitude and lowers the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together lowers it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which the grounding streams supply.

Widening or narrowing the temperature window does not remove the ill-conditioning of the solubility-only design. Varying ΔT over $\{120, 80, 40, 20, 10\}$ K moves the solubility-only conditioning through 1.7×10^3, 5.6×10^3, 2.3×10^5, 6.5×10^9 and 1.9×10^{16}, and the CRLB on ΔH_{fus} through 1,348, 2,249, 19,895, 3.2×10^6 and 8.0×10^6 J/mol. The point value quoted in the main text is the $\Delta T = 40$ K entry of that sweep and is not a design-independent constant. Multi-temperature data alone therefore does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps only partially: it reduces sd(ΔH_{fus}) to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic is not part of the primary argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated ($\text{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$). This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative) $\ln x_2$ fit residual, so their correlation tends to -1 as the fit improves, a permutation null that destroys any learned coupling still reproduces most of it, and the model's own factors barely co-vary. The large apparent crystal offset is likewise not a usable misgrounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics, and capping predicted T_m at a physical bound collapses the offset from ≈ -7.1 to ≈ -1.5.[1] We therefore base the non-identifiability claim on the reference-free Fisher/CRLB audit above, and present the compensation figures in the SI (§S5) as a cautionary diagnostic rather than a decomposition-quality measurement.

S6.3 Grounding the crystal factor: a dose-limited null

Whether the ceiling lies in the inputs rather than the closure can be tested by grounding a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the grounding-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE 45.0 → 46.6 K, a negligible change in the unfavourable direction). Downstream solubility did not improve either ($\ln x_2$ MAE 1.81 → 1.88; two nominally favourable but slight shifts, in $\ln x_2$ R^2 and the crystal offset $|\delta\Phi|$, do not amount to a gain). This result is *dose-limited and inconclusive rather than an unambiguous null.* The crystal grounding stream was applied here at a low dose, a small fraction of optimiser steps at low loss weight, so at the selected checkpoint the model had completed less than one full pass over the external pool. A stream that does not lower the MAE of the property it directly supervises cannot justify a strong conclusion. The result shows that *this low-dose intervention* does not move the ceiling, not that external crystal labels cannot. The activity-path $B_{\text{closure}}/B_{\text{insuff}}$ decomposition is uninformative here: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$, not a misspecified map, so the flat T_m recovery is more consistent with the crystal/activity split's structural non-identifiability (Lemma S1) and the encoder's transfer limitation (§S6.1) than with closure dominance. A properly dosed crystal-grounding test is left to future work.

S6.4 Chemistry-resolved structure of the asymptotic physics penalty

Scope. The two arms compared in this section differ from each other in capacity and in the per-arm optimisation settings of §2.1, and no hyperparameter-matched control was run, so nothing here is an accuracy claim about the thermodynamic bottleneck: Table 4 records that ΔMAE as a difference between separately tuned pipelines, and what follows is a hypothesis-generating description of where two particular pipelines differ (§3.6). "Physics penalty" names that measured difference throughout, not an effect of the bottleneck.

Across the full test set the two arms are separated by little more than label noise. On

[1] Permutation null (learned coupling destroyed) -0.90 to -0.93; model-factor $\text{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$; the Joback reference predicts $T_m > 1000\,\text{K}$ for a substantial fraction of these polycyclics.

the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.03, while the grounded physics arm (learned-σ COSMO-SAC closure) reaches 1.85 ± 0.05, for a global ΔMAE = MAE(physics) − MAE(control) = +0.14 (positive meaning the closure adds cost).[2] At this granularity the difference between the two pipelines is small and near label noise. Once the error is resolved along chemical axes, that difference is an average over regimes that disagree in sign.

The solvent axis is the most pronounced. Binning test pairs by solvent class and recomputing ΔMAE per seed (Table S17) shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln \gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class weakens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the seed-robust stratum in which the penalty reverses sign ($\Delta\text{MAE} = -0.22 \pm 0.11$ over seeds). This is exploratory rather than a confirmed improvement: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the asymptotic physics penalty shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved counterpart of the synthetic closure-fidelity sweep (§S11).

The solute axis shows a compatible but weaker pattern (Table S18). The closure costs most on ordinary neutral organics—oxygenated solutes, polyaromatics, and heterocycles—while the charged, high-$|\ln \gamma_2|$ stratum of salts is neutral. On this axis the penalty therefore does not track $|\ln \gamma_2|$: it falls on neutral organics, not on the ionic tail.

The novelty axis inverts the usual coverage expectation: the physics arm is neutral on the most novel chemistry (nearest-train Tanimoto ≤ 0.3) and hurts most on the most familiar (> 0.8; Table S18). This is exploratory rather than a confirmed effect: the seed spread is not a within-bin sampling interval over which so-

Table S17: Per-solvent-class asymptotic physics penalty ΔMAE = MAE(physics) − MAE(control) on the scaffold split (mean ± sd over 3 seeds; positive means the closure adds cost).

Solvent class	ΔMAE (mean ± sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

Table S18: Per-solute-class and novelty-stratum asymptotic physics penalty ΔMAE = MAE(physics) − MAE(control) on the scaffold split (positive means the closure adds cost). Entries carry the 3-seed spread where it was computed: the two Tanimoto strata and the charged/salt class. The other four solute classes are point estimates.

Solute stratum	ΔMAE
Oxygenated	$+0.32$
Polyaromatic	$+0.27$
Heterocycle	$+0.17$
Halogenated-aromatic	≈ 0
Charged / salt	-0.04 ± 0.19
Tanimoto ≤ 0.3 (novel)	-0.04 ± 0.18
Tanimoto > 0.8 (familiar)	$+0.51 \pm 0.09$

[2]The per-seed ΔMAE spans +0.05 to +0.25, so the point estimate is an imprecise 3-seed mean; being computed on unrounded per-seed means, the rounded 1.70/1.85 differ by 0.15 rather than 0.14.

lutes fall in the stratum, and the Tanimoto binning is one of several strata families we examine (solvent class, solute class, similarity) without multiplicity control. Taken as a direction rather than a tested quantity, the inversion suggests that the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit. That misfit is masked on novel solutes by the large errors both arms already incur there, and exposed on well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual, which is the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Together, the maps indicate that the physics decomposition incurs cost where the closure is misspecified relative to the solvent chemistry, and that it can only become a net benefit in the narrow acceptor-dominated regime the closure was built to describe.

S6.5 The data-efficiency sweep, and what survives its confounds

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct factorisation should outperform a data-hungry black box. We test it directly, subsampling the training set *by solute* at fractions 0.05–1.0 and training both the physics-grounded model and DirectGNN on the identical subsample, evaluated on the fixed scaffold test (Table S19, seed 42). The physics arm is *worse at every fraction*, and against the hypothesis the gap is *smallest* at the least data (ΔMAE = +0.08 at 5%); its R^2 is negative throughout. We do not read the growth of the gap with data (+1.0 at full) as a data-scale effect, because the sweep's endpoint does not reproduce the headline comparison. Its physics arm carries the NRTL closure, and at fraction 1.0 it gives 2.98 against 1.97 (control) on the same scaffold test where the three-seed runs of Table S2 give 1.7950 against 1.7022: ΔMAE an order of magnitude apart

Table S19: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(direct)}$.

train fraction	MAE		ΔMAE	R^2	
(by solute)	physics	direct	(phys−dir)	physics	direct
0.05	2.28	2.20	+0.08	−0.09	+0.07
0.10	2.44	2.07	+0.37	−0.25	+0.15
0.25	2.93	1.81	+1.12	−0.75	+0.38
0.50	3.12	1.98	+1.14	−0.98	+0.22
1.00	2.98	1.97	+1.01	−0.79	+0.22

(+1.01 here, +0.09 there; the second is taken from the unrounded means, since the rounded column would give +0.10, §S3). Both arms are worse here than their headline counterparts and the physics arm much the more so (+1.18 against +0.27 MAE), so the trend measures how the light per-fraction budget interacts with training-set size at least as much as how the physics prior does. The sweep is also single-seed, its physics-arm MAEs are non-monotone, and its two arms differ from each other in capacity and in the per-arm optimisation settings of §2.1, and both are wider than the headline arms (hidden width 256 and 128 here, 64 there). What survives is the level and not the trend (Table S19, plotted in Fig. S5)—the physics arm is worse at every fraction under a matched light budget—which is *consistent with* the sign rule (Prop. 1) rather than a sharp confirmation of it. The one inverting metric is top-1 solvent ranking (0.42 vs 0.23 at 5%; the ranking-versus-accuracy split of §S6.1), which does not translate into lower error. This is the last regime in which a physics prior should plausibly help; that it does not is consistent with the broad accuracy null and with the structural account—a non-identifiable split through a misspecified closure—as the operative explanation.

S6.6 Diagnostics that do not work, and why

The drift is not the closure's signed first-order inverse—but the test is under-resolved. A candidate account of the compensating drift is the closure's linear-response inverse, $\hat{\sigma} - \sigma \approx J^{+}(m - g(\sigma))$ (with $J =$

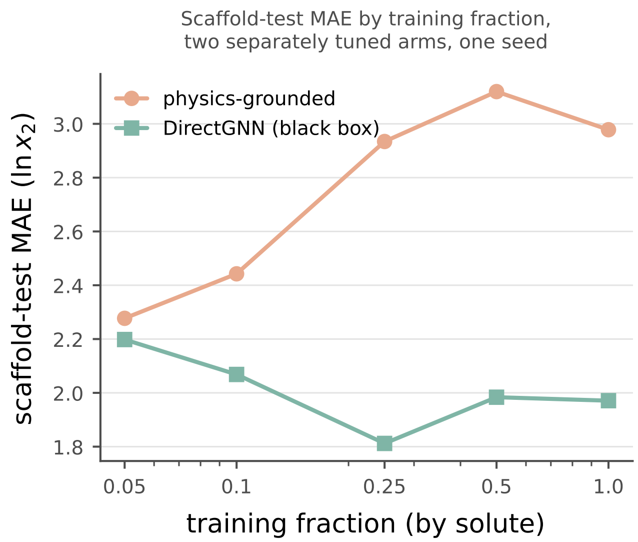

Figure S5: Data-efficiency curve: physics-grounded arm (salmon) and DirectGNN (teal) as a function of training-data fraction. *Scope:* one seed, five fractions, under a light per-fraction budget whose endpoint does not reproduce the headline comparison; the physics arm's MAEs are non-monotone. *Caveat:* the two arms differ in capacity and in per-arm optimisation, so nothing drawn here is attributable to the thermodynamic bottleneck, and this paper makes no accuracy claim in either direction (§3.6, §S6.5).

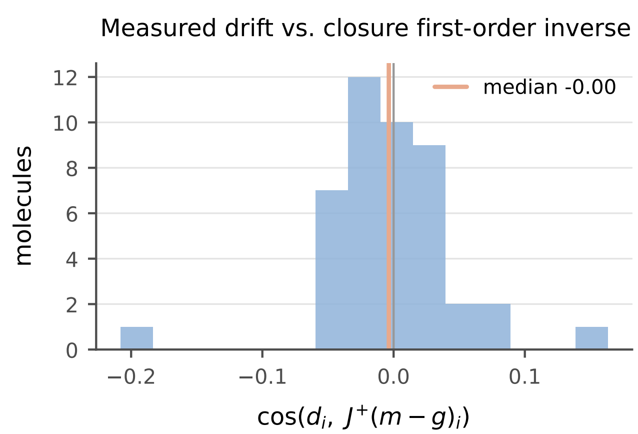

Figure S6: Per-molecule cosine between the measured drift $\hat{\sigma} - \sigma$ and the first-order closure compensation $J^{+}(m-g)$; the distribution is centred on zero (median -0.00).

$\partial g/\partial\sigma$), predictable a priori from J before any solubility training. On the matched set the per-molecule drift shows no signed alignment with $J^{+}(m-g)$ (median cosine -0.00; Fig. S6).[3] We read this as a signed negative only, and do not build on it, for two reasons intrinsic to the test. First, the matched set is sparse (median one pair per molecule), so each per-molecule Jacobian subspace is one-dimensional and the dimension-matched row-space controls are uninformative—a resolved direction common across molecules makes a molecule's "own" and "other" subspaces coincide. Second, and more fundamentally, the single-pair $J^{+}(m-g)$ is a poor *estimator* of the first-order compensation: the activity Fisher is near rank-one (PR $\approx$ 1.4 of 51, §3.3), so the closure's first-order inverse is ill-posed along all but one direction. A null here is therefore uninformative about whether the drift is first-order, not evidence that it is not. The load-bearing claim is the under-determination of σ by activity (§3.3), not the geometry of the drift.

The instrument behind §3.3—the rank of the activity Fisher $J^{\top}J$ ($J = \partial \ln \gamma_2^{\infty}/\partial\sigma$), read as a participation ratio $\mathrm{PR} = (\sum_i \lambda_i)^2/\sum_i \lambda_i^2$—is computationally inexpensive and oracle-free, which is its appeal for probing any physics-grounded latent. It is also easy to misread in three ways that inflate a null or a rank in the favourable direction without any warning sign; we record them because the corrections are not self-evident.

(i) Finite differences inflate the participation ratio. PR is a *tail* statistic, governed by the small eigenvalues. A closure evaluated by an unrolled fixed point carries a small, step-*independent* gradient noise (from the convergence tolerance, not truncation), so central finite differences lift the exactly- and near-zero eigenvalues onto a floor and PR rises. On the differentiable COSMO-SAC-2002 layer, autograd and finite differences on *identical* profiles give PR = 1.00 and 1.25—a 25% inflation from noise alone, though the per-gradient cosine is 0.995 (the direction is right, the tail is not). It

[3]The residual COSMO-SAC closure runs a segment-activity fixed point, so its autograd Jacobian is numerically unstable ($\|\partial g/\partial\sigma_{\mathrm{solvent}}\| \sim 10^{15}$ at infinite dilution); J is computed by finite differences on the converged closure.

is worse on a typed (3×51) profile basis. A closure that ignores the typing (2002) has 102 exactly-redundant coordinates that finite differences decorrelate into a spurious floor, inflating its PR from 1.0 toward 2.0, while a closure that uses the typing (2010) has no such redundancy. The same step therefore inflates the two *asymmetrically* and can manufacture a rank ordering with no physical content. *Autograd is the appropriate instrument where the closure is differentiable; with only finite differences,* PR *is interpretable against a known-rank or permuted control, never absolutely.*

(ii) "Own vs. other" controls are powerless when the resolved direction is common. A dimension- and smoothness-matched control aligns a molecule's drift against its own predicted compensation and against other molecules'. It has power only if the predictions differ. When the aggregate Fisher is rank-one (§3.3), however, the resolved direction is *common*, so "own" and "other" point almost the same way and the diagonal must sit inside the null—$p \approx 0.5$ regardless of the truth. On the matched set sparsity compounds this: a median of one pair per molecule makes each per-molecule Jacobian subspace one-dimensional, so "own" and "other" subspaces coincide exactly. *A control's null is interpretable only once its power is established: if the compared predictions have median cosine near one, the test cannot reject.*

(iii) Energy-in-subspace baselines need the effective rank and a cluster null. A random zero-mean vector in D dimensions places $k_{\mathrm{eff}}/(D-1)$ of its energy in a subspace of *effective* rank k_{eff} (the participation ratio of its singular values), which for near-collinear Jacobian rows is far below the nominal row count. On the matched set the per-molecule subspace is effectively one-dimensional, so the baseline is $\approx 2\%$ and the observed 5.5% is a two-to-threefold *enrichment*, not the "no concentration" a raw own-versus-other equality suggests: own equals other because both one-dimensional subspaces are the same common direction, not because the drift avoids the resolved one. With that direction common, N enriched molecules are not N independent draws—the enrichment needs a cluster null (§3.2), without which its significance is over-stated by $\sqrt{N/N_{\mathrm{eff}}}$. We therefore report the 5.5% as evidence in neither direction. *The baseline must be set by effective rank, and the null clustered by the shared direction.*

Each failure converts a spectral concentration statistic into a structural claim it does not support. The surviving statements of §3.3 avoid that conversion by carrying their null model and a power check inline.

S7 A second chemical closure: pKa through a Hammett LFER

The measured result of this section is stated in §3.5; what follows is the construction, the estimator and the sweep behind it. The instrument is not specific to COSMO-SAC. In a second real domain—aqueous pK_a of substituted aromatics—the fixed closure is a Hammett linear-free-energy relationship $g(\sigma_{\mathrm{H}}) = \mathrm{p}K_{a,0} - \rho\,\sigma_{\mathrm{H}}$ over the Hammett substituent constant σ_{H}, written with its subscript throughout so that it cannot be confused with the σ-profile of §2.1, with Hansch–Leo tabulated constants as the external oracle. The two classic LFER failure modes fall cleanly onto the two channels of Lemma 1: *resonance saturation* (a curvature the linear closure cannot represent) lands in B_{closure}, while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. On the OPERA experimental compilation[S15]—the canonical full set under a uniform, closure-independent quality filter that drops pre-ionized microstates (§S8)—this yields a *measured* fidelity sweep. The clean meta/para series (n=158) is well specified (RMSE 0.55, B_{closure} below the insufficiency floor). The ortho/heteroaromatic regime (n=846) degrades to RMSE 2.26, of which a within-scaffold estimator places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure. That estimator is an out-of-fold residual, so both figures are one-sided, and the a-priori channel assignment—ortho information the tabulated constant lacks—is consistent with the measurement rather than established by it (§S8); the sweep is the near-exact-to-degraded transition. The decisive test is the same oracle swap as on solubility—the learned $\hat{\sigma}_{\mathrm{H}}$ replaced by the tabulated $\sigma^{\star}_{\mathrm{H}}$ in the same closure, the sign of the change recorded per stratum—measured *in-distribution* across K=20 within-scaffold splits. It *flips*. On the clean meta/para pole feeding the reference constant *helps* (learned MAE 0.48 vs oracle 0.315; oracle better in 19/20 splits and in three of the four reaction series): the ceiling there is the noisy learned estimate, not the closure—a real-data pole on which grounding helps, and an internal positive control. On the ortho pole the same swap *hurts* (learned 0.83 vs oracle 1.62; oracle worse in **20/20** splits and in all four series, gap 90% CI $[+0.57, +0.95]$): the freely learned $\hat{\sigma}_{\mathrm{H}}$ absorbs the ortho/steric information the tabulated constant lacks. What reverses here is the sign of the grounding penalty; it reverses through the insufficiency channel—consistent with the one-sided bound above, though not isolated by it—rather than through the closure-error cancellation of §3.3: a second instance of the sign rule, not of that mechanism. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the accompanying bootstrap resolves nothing finer than series-level sign agreement—read as a sign test on four units, p=1/16 on the ortho pole and $p\approx$0.31 on the clean one—and is quoted as such (§S8).

Both signs obey one *crossover law*—grounding helps a pole iff the learned arm's error there exceeds the fidelity-set oracle error—which a competence sweep confirms (§S8). The well-specified pole's threshold ($\approx$0.315) is never beaten by the learned arm, so grounding helps at every training budget. The degraded pole's threshold (1.62) is high, so the learned arm beats it, and grounding hurts, across the whole interpolation range (down to 12% of the data), reverting to helps only under scaffold *extrapolation*, where $\hat{\sigma}_{\mathrm{H}}$ breaks (2.6$\pm$2.0) back above the threshold. The flip is thus a within-distribution, fidelity-driven crossover whose one scope boundary is predicted by the same law. It is the sharp, two-sided form of the penalty sign rule (Prop. 1); its "a fitted intermediate can beat the true one" half is known—plug-in efficiency under correct specification,[S8] pseudo-true fitting under misspecification[S16]—sharpened here into a fidelity-set threshold. The trained physics-vs-DirectGNN comparison is weaker and is not load-bearing (§S8). On an independent chemical closure the instrument separates a pole where grounding helps from one where it hurts, each with a sign consistent across splits and series; the full construction, the estimator validation and the measured sweep are in §S8 (Fig. S7).

S8 The pKa closure: full construction, measured sweep, and trained comparison

This section gives the full construction behind §S7: a fixed Hammett linear-free-energy relationship (LFER) $g(\sigma_{\mathrm{H}}) = \mathrm{p}K_{a,0} - \rho\,\sigma_{\mathrm{H}}$ over the Hammett substituent constant σ_{H} (with $\mathrm{p}K_{a,0}, \rho$ the per-scaffold Hansch–Leo–Taft reaction-series constants), for which Hansch–Leo tabulated constants provide an external oracle $z^{\star}$. The analogy is close: as with the σ-profile through COSMO-SAC, a learnable physical descriptor is pushed through a fixed, approximate, differentiable map. The two classic LFER failure modes fall cleanly onto the two error channels of Lemma 1: *resonance saturation*—a curvature in σ_{H} that the linear closure structurally cannot represent—is a mis-map of a function of $z^{\star}$ and lands in B_{closure}; *ortho/steric* field effects, absent from the tabulated constants, are information the intermediate lacks and land in B_{insuff}. The chemical scope is a fidelity sweep: meta/para monosubstituted benzoic acids, phenols, and anilinium ions are the near-exact ($B_{\mathrm{closure}} \approx 0$) regime, ortho (insufficiency-dominated) and polyfunctional the low-fidelity one.

This decomposition is directly visible before any training. Pushed through the fixed Hammett closure, p-nitrophenol reads p$K_a \approx$ 8.3 against an experimental $\approx$ 7.1 (and p-nitroaniline overshoots similarly): the linear closure is off by ~ 1 unit exactly where through-conjugation (σ_{H}^{-}) breaks it—B_{closure} observed in practice.

A controlled semi-synthetic construction on real tabulated σ_{H} (known ground-truth channels) validates the estimator and reproduces the paper's central distinction in this second closure (Fig. S7). Sweeping closure fidelity with a sufficient intermediate, the grounding gap $\mathcal{G}$ tracks B_{closure} (the plug-in estimate recovers the truth to within a few percent, e.g. 0.20 vs 0.21, 0.81 vs 0.81), the case where Theorem S1's proxy is valid. Sweeping the ortho/steric channel with a *well-specified* closure gives $B_{\text{closure}} = 0$ at every knob while $\mathcal{G}$ rises with the insufficiency fraction, from -0.035 at zero insufficiency through zero near a fraction of 0.25 to $+0.34$ at 0.7: at the three fractions where it is positive it is positive with B_{closure} identically zero, which is the pKa realization of the conflation that makes $\mathcal{G}$ an indicator rather than a certificate. The two negative points are the same statement in the other direction—a well-specified closure with a nearly sufficient intermediate leaves the oracle marginally ahead. The oracle-σ_{H} pipeline (scaffold-resolved substituent$\rightarrow \sigma_{\mathrm{H}}$ assignment) and this validation extend directly to real data. On the OPERA experimental pK_a compilation[S15] (7,904 records under the same net-neutral filter), applying the same decomposition to the fixed Hammett closure yields a *measured* fidelity sweep. On the clean meta/para series (n=158) the closure predicts experimental pK_a to RMSE 0.55, with the closure at the input-insufficiency floor: the leakage-immune lower bound MSE $- B_{\text{insuff}}^{\text{up}}$ is -0.2 (95% CI $[-0.4, -0.0]$), i.e. below zero, as a lower bound on the non-negative B_{closure} must sit when the LFER predicts as well as a flexible within-scaffold regressor. There B_{closure} is not separately resolvable and is consistent with a well-specified closure (per-scaffold systematic offsets ≤ 0.18). The ortho/heteroaromatic regime (n=846) degrades to RMSE 2.26, dominated by input insufficiency. Here the pKa latent is a *scalar* Hammett constant, so—unlike the solubility regime—B_{insuff} is estimated *directly* rather than through a coarsening of the closure's argument (§2.1.1): a within-scaffold out-of-fold regression estimator gives $B_{\text{insuff}} = 4.5$ (95% CI $[4.0, 5.1]$; $\approx$ 89% of the error—the tabulated σ_{H} carries no ortho information, exactly the a-priori channel assignment), leaving a closure misspecification $B_{\text{closure}} = 0.6$ ($[0.3, 0.8]$, a stratified within-scaffold molecule bootstrap). Directly estimated is not point-identified: the estimator is an out-of-fold residual mean square, which is an upper estimate of $\mathbb{E}[\mathrm{Var}(\mathrm{p}K_a \mid \sigma_{\mathrm{H}})]$, so 4.5 is an upper figure and the 0.6 it leaves is a lower one—one-sided in the same direction as everywhere else in this paper, and narrower only because the scalar latent needs no binning (§2.1.1, §S7). This is the near-exact-to-degraded transition, measured rather than asserted. This places the pKa closure near the phase-diagram boundary in its clean regime.

Data quality control (uniform, closure-independent). The trained runs use the canonical full OPERA set (7904 records). A small number of entries are supplied as pre-ionized microstates—an $[\mathrm{NH}^+]$-protonated aminopyridine at pK_a = -7.78, a pyrylium $[\mathrm{O}^+]$, among others—inconsistent with a neutral-scaffold Hammett assignment, and they degrade the clean pole (pyridinium RMSE $0.63 \rightarrow 3.83$). We drop them with one rule, *net formal charge* $\neq 0$, applied uniformly to both strata. It is closure-independent (it never references the oracle residual, so it cannot be circular) and keeps net-neutral substituents such as nitro $[\mathrm{N}^+][\mathrm{O}^-]{=}\mathrm{O}$. This removes 5 of 163 meta/para rows (restoring the clean pole to RMSE $\approx$0.55) and 352 of 1198 ortho rows (leaving 846; the ortho RMSE is essentially unchanged, so the filter does not manufacture the sign), for 1004 Hammett-amenable rows.

Evidence for the flip. We train the physics arm (encoder $\rightarrow$ $\hat{\sigma}_{\mathrm{H}}$ $\rightarrow$ fixed LFER) on K=20 stratified 80/20-within-scaffold splits—a Bemis–Murcko split collapses the meta/para pole to two ring systems and is unstable—and record the paired per-molecule error against the deterministic oracle, per stratum. Meta/para: learned MAE 0.48 ± 0.09 vs oracle 0.315 ± 0.07, gap -0.165, oracle better in 19/20 splits, 90% CI $[-0.28, -0.04]$. Ortho: learned 0.83 ± 0.07 vs oracle 1.62 ± 0.08, gap $+0.79$, oracle worse in 20/20 splits, 90% CI $[+0.57, +0.95]$ (both intervals from the scaffold-cluster bootstrap).

What the two bootstraps do and do not resolve. We report the sign statements above rather than the bootstrap P values, because neither bootstrap here carries the resolution a two-decimal P implies. The cluster unit is the Hammett reaction series, and only G=4 exist (benzoic acid, phenol, anilinium, pyridinium): the resampling law then has $\binom{7}{4}$=35 distinct outcomes and a finest atom $(1/4)^4 \approx 0.004$, so the cluster-bootstrap P(oracle worse)=1.00 on ortho is exactly the statement that all four series means favour the learned arm, and P=0.004 on meta/para that exactly one of them does, the other three favouring the oracle. Read as a sign test on four units those are p=1/16 and $p \approx 0.31$; the interval endpoints quoted above are percentiles of the same 35-outcome law and are no finer. This is the same small-G caution applied on the solubility side (§S3), where the P values are likewise read as a coarse band rather than as two-decimal quantities. The split bootstrap adds nothing independent: the 20 splits resample train/test assignments over the same molecules (each molecule lands in ≈4 test sets), so it prices split-to-split noise rather than molecule sampling, and with all 20 ortho gaps positive it returns 1.000 by construction—a restatement of the 20/20, not a second confirmation. What the flip rests on is therefore the sign consistency across splits and series and the size of the ortho gap (+0.79 against an oracle MAE of 1.62, ≈49%), not on a P value. A leave-one-series-out refit, the estimate that would settle the series-level question at G=4, was not run.

The crossover law and its scope boundary. A competence sweep (train fraction $\in \{1.0, 0.5, 0.25, 0.12\}$ and, as the low-competence extreme, scaffold extrapolation; six seeds each) confirms the sign is set by the crossover of the learned error against the fidelity-fixed oracle threshold. On meta/para the learned arm rises from 0.46 to 0.98 as data shrinks but never reaches the 0.31 oracle threshold, so grounding helps throughout. On ortho the learned arm stays below the 1.62 threshold across the whole interpolation range (0.85 at full data, 1.31 at 12%), so grounding hurts throughout; only scaffold extrapolation, where $\hat{\sigma}_{\mathrm{H}}$ breaks to 2.6 ± 2.0, pushes it back above the threshold and reverts the sign. The flip is therefore a within-distribution, fidelity-driven crossover, and the extrapolation reversion is predicted by the same law. A trained physics-vs-DirectGNN comparison agrees in direction overall but has weak absolute errors and an untuned control; the load-bearing second-domain results are the decomposition above and the sign-consistent flip.

S9 Data provenance and reproducibility

The broad IDAC set (n=477), and the UD profile database. The UD σ-profile database is the 2261-compound COSMO set reported by Fingerhut et al.[S10] and redistributed with the benchmark implementation of Ref.:[S1] 1941 compounds from the University of Delaware collection assembled in Sandler's group, which extends the freely available VT set with revised molecular conformations for about a third of its compounds, plus 320 generated by the authors of Ref.[S10] It ships both the Mullins-averaged single profile the 2002 kernel prescribes and the typed Hsieh three-profile set the 2010 kernel prescribes, with the cavity volume and, for most compounds, a tabulated dispersion parameter. The broad IDAC set is the intersection of that database with our expanded ThermoML infinite-dilution pull, at every temperature rather than only 298 K.

How the 14,900 *raw rows are extracted.* The pull walks the NIST ThermoML archive JSON records for four journals and keeps a row only where: the dataset has exactly two components; the property is an activity coefficient at infinite dilution, either declared so by a dataset

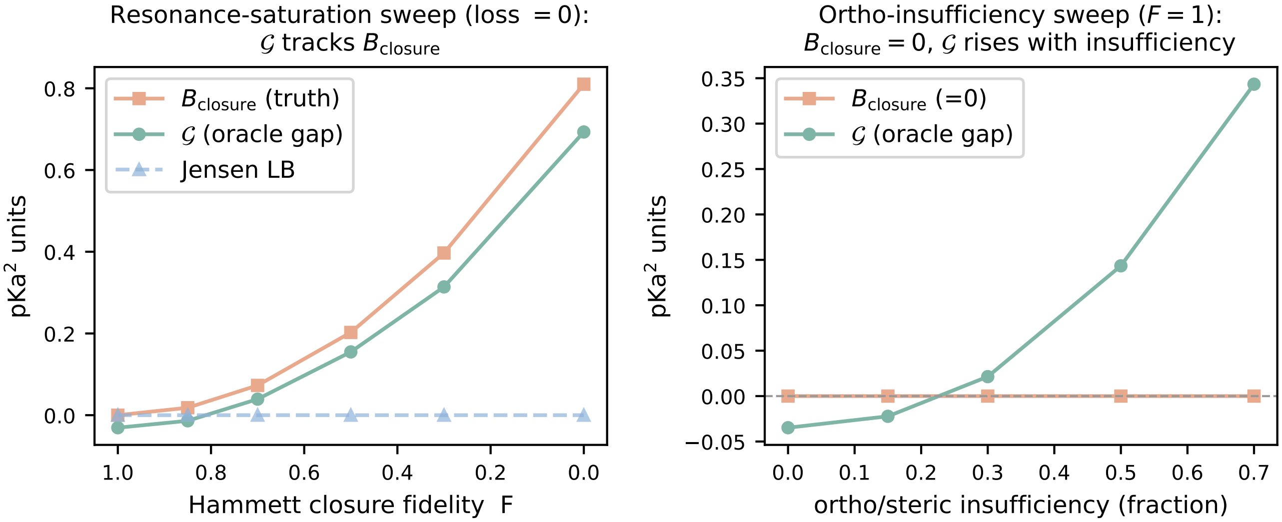

Figure S7: pKa through a Hammett LFER on a construction with known ground-truth channels. *Left:* closure fidelity varied with a sufficient intermediate; grounding gap $\mathcal{G}$, closure bias B_{closure}, and the assumption-light Jensen bound. *Right:* an ortho/steric insufficiency channel varied with a *well-specified* closure; $B_{\text{closure}} = 0$ at each knob while $\mathcal{G}$ rises with the insufficiency fraction, crossing zero near 0.25. Scope: §S8.

constraint or carried at a solute mole fraction of zero; exactly one component is the non-solute; a temperature is resolvable for the value; and γ^∞ is finite and positive. Components are mapped to SMILES through the record's own InChI. Duplicate (solute, solvent, temperature) records are then aggregated, with the record and DOI counts retained per row, which takes 14,910 extracted values to 14,900 rows. No solubility-side, method-side or quality-side filter is applied at this stage.

Construction ladder. 14,900 raw IDAC rows; 14,900 with a finite $\ln\gamma_2^\infty$ and temperature; 490 whose solute *and* solvent both resolve into the UD compound list by InChIKey (first-block fallback); 490 for which both the Mullins and the typed profile files exist and parse at the expected 51 and 153 bins; and 477 for which both components additionally carry a tabulated dispersion ϵ, which the *2010, dispersion on* arm needs and which we impose on every arm so that the three kernels are scored on identical rows. That leaves 78 solutes, 36 solvents and 35 distinct temperatures from 273.2 to 403.1 K, with $\mathrm{Var}(m) = 4.85$ and $\overline{m} = 3.42$—a harder and far less ideal regime than the VT-2005-matched set's $\overline{\ln\gamma_2^\infty} = 0.75$. It is not disjoint from the VT-2005-matched set: 57 of the 60 VT-2005-matched pairs appear in it at 298 K, evaluated there on UD rather than VT-2005 profiles. The two are therefore the same chemistry measured against two profile databases, not independent samples, and we never combine them. The 3 VT-2005-matched pairs it excludes are exactly the 3 that carry no tabulated dispersion ϵ: bromobenzene and iodobenzene in 1,2-ethanediol, and 2-bromopropane in pyridine. The "dispersion-defined 57" of §3.2.3 and the "57 shared with the broad IDAC set" are therefore one set and not two.

Why neither set can observe B_{insuff} directly. $B_{\text{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star, T)]$ would be point-identified only by two or more independent determinations at a common closure argument, and neither design supplies one. VT-2005 assigns exactly one profile per compound; the 44 matched compounds have distinct profiles (closest pair L_2=3.4 against a median profile norm of 44), and no pair carries a replicate, so on the VT-2005-matched set no two rows sit at a common $z^\star$. The broad IDAC set comes closer without reaching it: 359 of its 477 rows share a molecule pair with another row, but at a different temperature and hence at a different $(z^\star, T)$. On these two sets B_{insuff} is accordingly reached only as an upper bound—by coarsening in the law-of-total-variance cells, and out of fold in the fifteen fitted rows of Table S7—and B_{closure} correspondingly only as a lower bound; where the latent is a scalar the headline estimate is the out-of-fold residual, one-sided in the same direction (§2.1.1, §S8).

The matching rule, and its asymmetry with the VT-2005-matched set. The VT-2005-matched set is matched by canonical SMILES and this set by InChIKey with a 14-character first-block fallback, which is looser and, since a $z^\star$ imported across a tautomer, protonation or stereochemical variant inflates MSE and hence B_{closure}, looser in the anti-conservative direction. We therefore count it. Of the 90 distinct molecules, 86 resolve by full InChIKey; four resolve only through the first block (`C=CC1CC=CCC1`, `CCC(C)N`, `CC(Cl)CCl`, `CCC(C)C(C)=O`), touching 12 of the 477 rows. In all four the discrepancy is the stereochemical layer and not a tautomer or a protonation state: each query key ends `-UHFFFAOYSA-N`, an unspecified stereocentre, against a stereo-labelled UD entry with the same connectivity. On the 465 exactly matched rows (178 distinct pairs) the decomposition is unchanged, read at the headline cell of the grid (eight equal-count bins, unbiased within-bin variance, edge values to the upper bin), which is what recomputes these two lines from the deposit: deployed convention MSE 1.930, $B_{\text{insuff}} \lesssim 0.699$, margin $+0.53$ at $P \approx 0.9$; full convention 1.926, 0.495, $+0.94$ at $P > 0.95$.

Composition, and where the squared error sits. The set's chemistry is small in both channels: n-alkanes, aromatic hydrocarbons and ketones supply 243 of the 477 solute rows, and water and the four ethylene-glycol oligomers 293 of the solvent rows. Its overlap with this paper's solubility corpus is correspondingly asymmetric—only 0.46% of corpus rows carry a solute from this set, while 34 of its 36 solvents cover 59% of corpus rows—which is what the main text means by the set covering the solvent side of deployment and not the solute side. The error concentrates in the same places: under the deployed convention 45% of the squared error sits in the four ethylene-glycol solvents and 32% in the 37 rows carrying water as solute, and under the full convention 49% sits in those same four solvents. The fidelity null of §3.2.3 is therefore a small-molecule and glycol-heavy null, and the same concentration is what makes the aggregate margin a composition statistic (Table S8). Broken out by the classification of that table, three solvent classes holding 47% of the rows carry 74% of the squared error while the acceptor-only aprotics carry 0.6% of it on 12% of the rows, and by publication the three largest contributors—32%, 24% and 20%—carry 76% between them.

Design and stratification. The 477 rows are 185 distinct solute–solvent pairs: 118 pairs carry one row, 38 carry five or more, and the largest carries 15. Counted per row the design ratio is $\dim(z^\star, T)/n = 103/477 \approx 0.22$; counted per distinct pair, 0.56, against 1.7 on the VT-2005-matched set. Molecule-pair identity coarsens $(z^\star, T)$, so $\mathbb{E}[\mathrm{Var}(m \mid \text{pair})]$ is a valid B_{insuff} upper bound over the 359 rows that support it; it is 0.046, under 1% of $\mathrm{Var}(m)$ and thirteen times tighter than the binning bound on those same rows (0.61 deployed, 0.39 full). We report it and do not promote it, because all 67 multi-row pairs come from a *single* publication each: those rows are one laboratory's temperature series, so the bound omits the between-determination component of the superpopulation (§2.1.1) while the binning bound, whose eight bins mix all 15 sources, does not. It bounds a sub-component, so the reading that holds is that the binning bound is slack rather than that B_{insuff} is near zero. The same fact —that the temperature axis carries under 1% of $\mathrm{Var}(m)$ at a fixed profile pair—makes the choice between conditioning on $z^\star$ and on $(z^\star, T)$ numerically immaterial on this set, though only the latter is licensed by Lemma 1 and it is the one we use. Collapsing to one row per pair (each the row nearest 298 K) gives MSE = 1.679, $B_{\text{insuff}} \lesssim 0.495$ and a margin of $+0.69$ at $P > 0.95$ under the full convention (1.540, 0.635, $+0.27$, $P \approx 0.8$ deployed). By method, 334 rows are chromatographic, 37 chemical or heterogeneous equilibration and 106 carry no method field; by source, 15 publications in four journals (*J. Chem. Eng. Data* 203, *J. Chem. Thermodyn.* 114, *Fluid Phase Equilib.* 108, *Thermochim. Acta* 52), the largest single publication contributing 105 rows. On the 334 chromatographic rows alone, re-binned within the stratum, the activity error is MSE = 1.783 against $B_{\text{insuff}} \lesssim 0.36$ under the full convention—a margin of +1.06 ($P > 0.95$), wider than the full set's +0.95—and 1.474 against $B_{\text{insuff}} \lesssim 0.58$ under the deployed one, a margin of +0.31 ($P \approx 0.9$), narrower than the full set's +0.51. The cut therefore does not move the sign under either convention, and it does not uniformly widen the margin either. A DOI-clustered bootstrap of the full-set margin gives $P > 0.95$ with 90% $[+0.50, +1.57]$ (full) and $P \approx 0.9$ with $[-0.07, +1.74]$ (deployed), and a pair-clustered one $P > 0.95$ with $[+0.71, +1.27]$ and $[+0.23, +1.16]$ respectively. This is the stratification the 60-pair VT-2005-matched set cannot support, and it is why the broad IDAC set carries the map: the quality cut that *inverts* the enlarged 298 K set (§3.6, limitation (i)) leaves the sign alone here under either convention. What it does not buy is an aggregate verdict, for the reason Table S8 and §3.2.1 give: the chromatographic cut and the pair collapse each leave the sign standing, and the chemistry cut compounded with the pair collapse does not.

Sources. Solubility: BigSolDB 2.0,[S17] partitioned by Bemis–Murcko scaffold—molecules grouped by ring system plus linker, side chains stripped. Crystal $T_m/\Delta H_{\text{fus}}$: an open melting-point pool plus a group-contribution prior. True σ-profiles: the VT-2005 database, the Virginia Tech 2005 compilation of quantum-chemically computed screening-charge profiles for ∼1,400 compounds[S18] (ingested to `sigma_profiles.csv`, 1432 compounds, with the COSMO cavity volume V_{cosmo} for the combinatorial term). These are quantum-chemically computed for a single reference conformer and protonation state; mismatch to the experimental measurement conditions is noise in $z^\star$ that inflates B_{insuff}, but under a Berkson-type displacement it also loads onto B_{closure} through the curvature of g, so we treat it as an unsigned limitation and *not* as conservative for the closure-dominance verdict (§3.2). Concretely, if the true latent is $z^\star$ displaced by a mean-zero (Berkson-type) ε, then even a perfectly specified g has $B_{\text{closure}} = \mathbb{E}\big[(\mathbb{E}_\varepsilon[g(z^\star+\varepsilon)] - g(z^\star))^2\big]$, a pure curvature effect that vanishes only for affine g—and under classical rather than Berkson error, not even then. We do not bound that curvature share. Infinite-dilution activity coefficients: ThermoML, the NIST/IUPAC archive of experimental thermophysical data.[S19] Enlarging the reference σ-profile set would not materially grow the matched activity set: of the 298 K ThermoML IDAC pairs, only two solutes lack a VT-2005 profile while their solvent has one—the binding constraint is joint IDAC×profile coverage, not profile availability. What does enlarge the VT-2005-matched set is a broader extraction on the IDAC side: the expanded ThermoML pull released with the code matches 115 pairs at 298 K. We do not adopt it as a replacement set because its verdict is quality-cut sensitive: it widens the margin from 0.24 to 0.35 but *inverts* it to −0.13 on the 100 pairs labelled gas chromatography, and that sensitivity is unresolved at 298 K. It is not a verdict on the extraction, which matters because the broad IDAC set is built from the same pull: there the same cut leaves the sign alone under either convention (+0.51 → +0.31 deployed, +0.95 → +1.06 full). What differs is resolution, not provenance—at 298 K a 15-row cut inverts a margin already inside the design's resolution, whereas on 477 rows a 143-row cut moves one without inverting it—so we use the enlarged pull where it is stratifiable and keep the strict n=60 set where the paradox's own oracle is defined. The deployed VT-2005-matched set is also narrowly sourced: the curated IDAC compilation yields 74 VT-2005-matchable pairs under canonical SMILES (60 at 298 K plus 14 off-temperature), all by gas chromatography and all from three publications, 33 pairs from one, so neither a method-stratified nor a source-clustered re-estimate is available on it. The VT-2005-matched set is matched to VT-

2005 by RDKit canonical SMILES on solute and solvent, the rule the deployed σ-oracle itself uses; matching by InChIKey instead admits 85 pairs over 46 solutes and 19 solvents at 298 K in place of 60/41/17 (both counts are recorded in `decomposition.json`). We keep the stricter rule so that the $z^\star$ charged against each label is the profile the deployed pipeline would supply, rather than one imported across a tautomer or protonation variant—the same mismatch that makes the reference imperfect above. The decomposition was not re-estimated on the InChIKey-matched set; whether the verdict moves on those 25 extra pairs is untested. The broad IDAC set uses the looser InChIKey rule, so the two sets are matched under different standards; that asymmetry is quantified above (four molecules, 12 rows, verdict unchanged on the exactly matched subset) rather than left implicit. Second-domain pKa: the OPERA experimental compilation[S15] (`pKa_QR.sdf`, 7904 QSAR-ready records, public-domain US EPA; cited, not redistributed); the uniform net-formal-charge filter reduces this to the 1004 Hammett-amenable rows (158 meta/para, 846 ortho/heteroaromatic) analysed in §S8. The Bemis–Murcko scaffold split follows the main text. Because the seeded split is not stable across pipeline versions (§2.1), the numbers in this paper are computed on a *single, pinned* split, whose per-file SHA-256 each run's manifest records and whose files are deposited with the checkpoints rather than versioned—with the one run whose split provenance was not logged, recorded in the footnote.[4] Melting-point tables are auto-detected as °C or K to avoid a +273 K double-conversion.

Grounding-stream provenance, and one build that violates the guard. The σ-stream builder drops any pool molecule whose Bemis–Murcko scaffold appears in the validation or test split and aborts rather than proceed when a split file is missing; a correct build retains 1319 of the 1427 VT-2005 molecules that parse into the pool. The stream file was subsequently rebuilt with the exclusion pointed at the splits of a small debug dataset: it carries 1425 rows, of which 106 (99 train, 7 validation) have a scaffold in the real test $\cup$ validation union of 2702 scaffolds. The overlap is not confined to scaffold relatives: 91 of those 106 rows are themselves held-out solutes by canonical SMILES — 65 test solutes and 26 validation solutes — and the correctly guarded build leaves none of them. The crystal stream's build on disk records the correct 2702-scaffold exclusion, so among the streams feeding the runs reported here the defect is confined to the σ one. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. The rebuilt file postdates every prediction file behind Fig. 2 and Table S2 and predates the two single-seed oracle-application controls (§S3) and the output-residual arms (§3.4). That ordering is consistent with the headline arms having used the guarded build and the later controls the leaking one, but these are timestamps on local copies of cloud-run outputs and settle nothing on their own. Three facts bound what the leak can do. It is common-mode across the paradox contrast, the learned-σ and σ-oracle rows being one checkpoint evaluated two ways, so it cannot generate the gap between them; DirectGNN uses no grounding stream, so a leak can only favour the physics arm that loses to it; and on the n=44 matched set of §3.3 a correct guard would have removed one molecule (piperidine). What it does compromise is auditability, and with it the reach of the split guarantee: an arm trained on that build would have had reference σ-profiles — though no solubility labels — for 65 test solutes, and the guarantee stated in §2 is therefore a property of the released builder rather than of every stream file we trained on. Closing this requires hashing each stream build beside the split hash, which was not done for these runs.

The two decomposition sets, deposited row by row. Because the clo-

[4] The three-seed surrogate isolation (§3.3) was produced on a separate run whose per-run split provenance was not logged; its matched set is the same n=44 molecules, derivable from the committed split independently of the model, so it is consistent with the reported split. We record the split hash in the runner so future runs are auditable directly.

sure/insufficiency verdict rests on a newly constructed set, both sets are deposited with the Supporting Information as row-level tables that need no code to inspect. The results that otherwise require re-running the code are, specifically, the broad IDAC set with its design and provenance audit, the 2010 kernel's own-latent floors, the fitted kernel residual and the estimator grid. **Data Set S1** (`broad_idac_set_477.csv`) carries one row per measurement: pair identifier, solute and solvent SMILES, the UD compound key matched to each and which rule matched it (full InChIKey or first-block fallback), temperature, the experimental $\ln \gamma_2^\infty$, the *four* closure outputs—2002 and 2010-dispersion-off, each in the full and the residual-only combinatorial convention, so that the deployed-convention kernel contrast of §3.2.3 is recomputable from the file—the ThermoML method field, the source DOI and journal, and a flag for the 57 rows shared with the VT-2005-matched set. Some analysis artifacts label that same table `representative_set_477.csv`. **Data Set S2** (`vt2005_matched_set_60.csv`, printed in full as Table S20) carries the same fields for the 60 VT-2005-matched pairs, with three differences that follow from how the VT-2005-matched set is defined. Its closure outputs are the VT-2005 ones the deployed σ-oracle uses (columns `g_full`, `g_res`) alongside the four UD-profile outputs that make it commensurable with Data Set S1. Its matching rule is canonical SMILES throughout, so it carries neither the per-row match-rule columns nor the UD compound keys. It also names each compound in words as well as in SMILES. It additionally carries a per-row record count, 1 throughout. All 60 rows sit at 298 K, all are chromatographic, and their three source DOIs divide the set 33/20/7. The counts, margins, stratified re-estimates and clustered bootstraps reported for either set are recomputable from these two files with the estimator definitions of §3.2; what they do *not* restore is the provenance of the closure evaluations themselves, which still requires the code.

Compute ledger. The headline σ-grounding comparison (Fig. 2, Table S2) is a full-corpus GPU run (three seeds); the closure/insufficiency decomposition (§3.2), the Fisher audit (§S6.2), the noise floor, the encoder probe, and the temperature and mechanism diagnostics of §S5 are CPU-reproducible from deposited artifacts. The prediction CSVs, the model checkpoints and the larger analysis artifacts will be deposited separately, in a Zenodo archive **[whose DOI is registered at submission]**, rather than in the repository. The broad IDAC set's design and provenance audit—distinct-pair counts, within-pair variance across temperature, pair-blocked out-of-fold estimators, method and source-DOI composition, the method-stratified and cluster-resampled margins, the kernel contrast split into the shared 57, the complementary 420 and the 80 complementary rows that are also at 298 K, the VT-2005-matched floor recomputed on UD profiles, and both sets' floors under the 2010 kernel's own typed latent—regenerates in one CPU run of `scripts/analysis/run_b_insuff_representative_audit.py` into `results/b_insuff/representative_audit.json`. A second CPU run, `scripts/analysis/run_b_insuff_convention_audit.py` into `results/b_insuff/convention_audit.json`, produces the convention-rule material: both sets' margins and clustered bootstraps under each combinatorial convention, the per-cell broad-set grid of Table S4, the InChIKey match-rule audit and its exactly matched re-estimate, the pair-conditional bound and its single-source diagnosis, the typed latent's dimensions, and the two deposited row-level tables. Those files are among the artifacts limitation (x) records as uncheckable until the repository is released; the two deposited tables are the part of that material which can be checked without it.

Seeds and determinism. The controlled comparisons of §3.1 use seeds $\{42, 43, 44\}$ and are reported on a cross-arm intersection lock (rows supervised and finite in every arm); the surrogate isolation (§3.3) uses seeds $\{0, 1, 2\}$; the training-time and channel-swap oracle con-

Table S20: **Data Set S2** (`vt2005_matched_set_60.csv`) in full: the VT-2005-matched set (n=60), row by row, deposited so the set can be inspected without running code. Read in two blocks of 30, left then right.

#	solute[a]	solvent	m	g_{full}	g_{res}	broad	#	solute	solvent	m	g_{full}	g_{res}	broad
1	`Brc1ccccc1`	`OCCO`	3.81	2.33	2.40	n	31	`CCc1ccccc1`	`c1ccncc1`	0.36	0.35	0.39	y
2	`C#CCCCCC`	`c1ccncc1`	0.44	-0.42	-0.34	y	32	`CO`	`C1CCNCC1`	-0.84	-2.38	-2.21	y
3	`C1=CCCCC1`	`c1ccncc1`	0.91	0.75	0.75	y	33	`CO`	`CC(C)(C)N`	-1.24	-3.15	-2.99	y
4	`C1CCOC1`	`c1ccncc1`	-0.07	0.03	0.03	y	34	`CO`	`CCC(C)N`	-0.97	-3.20	-3.02	y
5	`C=CCCCCC`	`c1ccncc1`	1.27	0.93	1.03	y	35	`CO`	`CCCCCCN`	-0.97	-3.23	-2.87	y
6	`CC#N`	`c1ccncc1`	0.38	0.25	0.30	y	36	`CO`	`CCCCN`	-1.05	-3.00	-2.79	y
7	`CC(C)(C)N`	`CO`	-1.39	-2.07	-1.85	y	37	`CO`	`CCCNCCC`	-0.37	-2.33	-1.97	y
8	`CC(C)(C)N`	`c1ccccc1`	0.28	0.39	0.38	y	38	`CO`	`CCNCC`	-0.67	-2.35	-2.15	y
9	`CC(C)=O`	`c1ccncc1`	0.17	-0.13	-0.12	y	39	`CO`	`Cc1cccc(C)n1`	-0.49	-0.78	-0.49	y
10	`CC(C)Br`	`c1ccncc1`	0.30	0.22	0.21	n	40	`CO`	`Cc1ccccn1`	-0.24	-1.07	-0.86	y
11	`CC(Cl)CCl`	`c1ccncc1`	-0.07	-0.05	-0.05	y	41	`CO`	`Cc1cccnc1`	-0.06	-1.06	-0.85	y
12	`CCC(C)N`	`CO`	-0.29	-1.99	-1.71	y	42	`CO`	`Cc1ccncc1`	-0.39	-1.11	-0.90	y
13	`CCC(C)N`	`c1ccccc1`	0.14	0.42	0.42	y	43	`CO`	`NC1CCCCC1`	-1.11	-3.21	-2.98	y
14	`CCCCCCC`	`c1ccncc1`	1.91	1.60	1.71	y	44	`Cc1ccc(C)cc1`	`c1ccncc1`	0.44	0.39	0.45	y
15	`CCCCCCCC`	`c1ccncc1`	2.07	1.74	1.91	y	45	`Cc1cccc(C)c1`	`c1ccncc1`	0.43	0.37	0.43	y
16	`CCCCCCCCC`	`c1ccncc1`	2.17	1.86	2.12	y	46	`Cc1cccc(C)n1`	`CO`	-0.06	0.14	0.64	y
17	`CCCCCCCCCC`	`c1ccncc1`	2.26	1.97	2.32	y	47	`Cc1ccccc1`	`OCCO`	4.53	2.61	2.67	y
18	`CCCCCCCCCCC`	`c1ccncc1`	2.32	2.08	2.52	y	48	`Cc1ccccc1C`	`c1ccncc1`	0.34	0.33	0.37	y
19	`CCCCCCN`	`CO`	2.62	-2.35	-1.66	y	49	`Cc1ccccn1`	`CO`	-0.29	-0.21	0.12	y
20	`CCCCCCN`	`c1ccccc1`	2.79	0.51	0.59	y	50	`Cc1cccnc1`	`CO`	-0.20	-0.36	-0.03	y
21	`CCCCCOC(C)=O`	`c1ccncc1`	0.11	0.09	0.26	y	51	`Cc1cccnc1`	`c1ccccc1`	-0.04	-0.01	-0.01	y
22	`CCCCCl`	`c1ccncc1`	0.43	0.37	0.38	y	52	`Cc1ccncc1`	`CO`	0.00	-0.41	-0.08	y
23	`CCCCN`	`CO`	-0.78	-2.18	-1.85	y	53	`Clc1ccccc1`	`OCCO`	3.69	2.29	2.35	y
24	`CCCCOC(C)=O`	`c1ccncc1`	0.14	0.04	0.14	y	54	`Fc1ccccc1`	`OCCO`	3.47	2.03	2.06	y
25	`CCCN(CCC)CCC`	`CCCCCCCCCCCCCCCC`	0.01	-0.03	0.07	y	55	`Ic1ccccc1`	`OCCO`	3.74	2.45	2.54	n
26	`CCCN(CCC)CCC`	`CO`	2.17	1.96	3.15	y	56	`NC1CCCCC1`	`CO`	1.57	-2.17	-1.81	y
27	`CCCN(CCC)CCC`	`c1ccccc1`	1.03	0.54	0.79	y	57	`NC1CCCCC1`	`c1ccccc1`	1.86	0.45	0.46	y
28	`CCCNCCC`	`CO`	-0.09	-0.31	0.40	y	58	`c1ccc2ccccc2c1`	`OCCO`	5.47	2.71	2.85	y
29	`CCCNCCC`	`c1ccccc1`	0.39	0.28	0.35	y	59	`c1ccccc1`	`OCCO`	3.47	2.11	2.12	y
30	`CCNCC`	`CO`	-1.11	-0.64	-0.30	y	60	`c1ccncc1`	`CO`	0.18	-0.28	-0.09	y

[a] Columns: m, the experimental $\ln \gamma_2^\infty$ at 298 K; g_{full} and g_{res}, the VT-2005-input COSMO-SAC-2002 output under the full and residual-only conventions; *broad* (the deposit's `in_broad_idac_set`), the 57 pairs that also appear in the broad IDAC set, the three that do not being the bromine- and iodine-containing pairs of §3.2.3. The 477-row broad IDAC set is the companion machine-readable Data Set S1.

trols, the output-residual arms (§3.4), the data-efficiency curve (§S6.5) and the loss-simplification ablation (§S1) are seed 42 only. The decomposition is seed-independent (its closure has no learnable parameters). One artifact caveat: a save-time bug truncated the released per-row prediction file for the seed-44 oracle solubility arm to 295 rows, a small fraction of the test set. The three-seed headline summaries (Fig. 2, Table S2) were aggregated at run time, before that file was written, so they are unaffected; only the one on-disk artifact is partial, and any per-row re-analysis of the solubility arms therefore uses the two complete seeds (42, 43). The learned-σ and DirectGNN solubility files and the activity-level (n=60, n=44) artifacts are complete for every seed.

S10 Black-box probe programme: full tables

All three studies use the same current-split DirectGNN checkpoints (seeds 42/43/44, test MAE 1.749/1.674/1.684) whose manifests pin the train/val/test SHA-256 to the split on disk. The probed object throughout is the control's 788-dimensional pair representation. Every number here is CPU-computed; none required a GPU.

Crystal-term decodability (Table S21). Probe fitted on 152 solutes / 14,348 rows, scored on 13 held-out solutes / 527 rows, solute overlap zero. Φ^* is 92.2% between-solute variance, so the effective n is 13 clusters.
Controls: positive—model's own prediction R^2 = 0.980, temperature R^2 = 0.9997;

Table S21: Crystal-term decodability per seed, with the permutation null, the drop-one-solute jackknife and the leverage carried by the worst solutes.

seed	R^2(model)	perm. p	jackknife range	sign flips	top-3 leverage
42	−0.506	0.894	[−1.151, −0.228]	0/13	56.2%
43	+0.111	0.055	[−0.286, +0.333]	3/13	59.8%
44	+0.027	0.177	[−0.253, +0.237]	6/13	57.2%

negative—random target −0.0006, molecule-blind constant profile −0.015, raw RDKit descriptors −0.514 (set A) and −0.928 (set B). Null $p_{95} = +0.105/+0.117/+0.096$ and max $= +0.316/+0.380/+0.346$ over 1000 draws.

Two features of Table S21 bear on how such probes should be read. First, the between-seed spread (0.616 range, sd 0.334) exceeds every individual value: a single-seed run would have been unreadable, and seed 43 alone would have read as a near-miss. Second, a decision rule of the common form $R^2(\text{model}) - R^2(\text{raw}) \geq 0.10$ would have declared success here on all three seeds under descriptor set B—including seed 42, whose probe is worse than a molecule-blind constant—and on two of three under set A. The subtrahend is an unregistered analyst choice worth more than the effect; we report R^2(raw) as a level beside R^2(model) and never as a difference.

Temperature structure (Table S22). Slopes are fitted at fixed (solute, solvent); a slope pooled across solvents would confound activity differences with temperature. Ceiling, measured on the data before any model: $r = +0.239$ on train (1353 pairs / 151 solutes) and −0.125 held out (66 pairs / 13 solutes, permutation $p = 0.581$).

Organisation of the representation (Table S23). Excess over a permutation null matched to the factor's level, on the full test set.
As a share of the solute-identity ceiling: scaffold 100.3/99.9/99.8%, $\log P$ 13.6/8.2/4.7%, TPSA 9.0/10.3/7.1%, MolWt 5.3/2.3/3.1%. That 99.8–100.3% scaffold range is the organisation measure §3.6 points here for. The scaffold share is pinned near one by the split rather than by the representation: of the 124 scaffolds spanned by the 147 test solutes only 11 hold more than one solute, so only 34 solutes share a Bemis–Murcko scaffold with another, carrying 492 of the 5608 rows (8.8%), and the two partitions coincide on 91% of the data. The raw variance shares run the other way (seed 42: scaffold 0.636 against solute 0.639), so the *raw* ratio falls short of one by 0.4/0.7/0.9% across the three seeds, and it reaches one only through the differing size-matched permutation nulls (0.022 against 0.026); no interval is attached to that 0.4–0.9% gap. The row therefore bounds what the representation adds beyond scaffold and does not measure within-scaffold discriminability (§3.6). A direct measure on the 34 solutes that do share a scaffold sizes what it bounds: two same-scaffold solutes in a common solvent sit at 0.21–0.40 of the between-scaffold representation distance across the three seeds (112 pairs over 11 scaffolds; 90% scaffold-cluster intervals inside $[0.14, 0.57]$). Molecules sharing a scaffold are compressed rather than unresolved.

Temperature is decodable from this vector at $R^2 = 0.9997$ yet occupies only 3–4% of its variance (rows above): decodability of a quantity is not evidence that a representation is organised around it, which is why the crystal-term probe is run as a held-out transfer test rather than an in-sample decode.

On the 331 rows carrying a measured Φ^*, Φ^* accounts for 18.3/42.5/18.0% of that subset's own solute-identity ceiling. This does not contradict the held-out null above. Φ^* is a solute-level quantity and the representation is organised by solute identity, so any solute-level variable shares variance with it in sample; the decodability probe above asks the different and sharper question of whether the mapping transfers to solutes the probe never saw. In sam-

Table S22: The model's temperature response against the empirical slope and against $-\Delta H_{\mathrm{fus}}/R$, with the curvature of both in $1/T$.

	seed 42	seed 43	seed 44
corr(model, empirical slope)	+0.294	+0.406	+0.195
corr(model, $-\Delta H/R$)	+0.066	−0.283	−0.028
median |model − empirical| slope	1329	1088	1007
curvature in $1/T$ (data: 6.8×10^5)	2.0×10^6	2.1×10^6	2.3×10^6

Table S23: Share of representation variance per organising factor, as an excess over a permutation null matched to the level at which that factor varies.

factor	seed 42	seed 43	seed 44	null level
solute identity (147 levels)	+0.613	+0.587	+0.597	row
solute scaffold (124 levels)	+0.614	+0.586	+0.595	row
scaffold beyond arbitrary solute grouping	+0.088	+0.082	+0.082	solute
solvent identity (70 levels)	+0.278	+0.301	+0.271	row
temperature	+0.040	+0.029	+0.037	row
$\log P$ (solute)	+0.083	+0.048	+0.028	solute
TPSA (solute)	+0.055	+0.061	+0.042	solute
MolWt (solute)	+0.032	+0.013	+0.018	solute

ple the organisation is partly Φ^*-aligned; out of sample that alignment does not carry.

S11 Closure fidelity sets the reference-input penalty: a controlled synthetic family

To isolate the mechanism we construct a controlled synthetic experiment in which only the closure's fidelity varies (input quality, sample size, and estimator held fixed). A teacher draws a latent $z^\star$ and a target through a known map T; a fixed candidate closure g_F reconstructs it under a tunable misspecification of scalar fidelity F ($F = 1$ recovers T exactly, smaller F pushes g_F away). The reference-input penalty $\Delta R^2 = R^2_{\mathrm{oracle}} - R^2_{\mathrm{head}}$ is a measure of B_{closure} in this construction—the free head recovers $\mathbb{E}[m \mid z^\star]$ while g_F is the misspecified map. Because ΔR^2 *equals* B_{closure} by design here, the sweep verifies that the instrument responds to fidelity as constructed; it is a check on the construction, not an independent test of the hypothesis. As F falls the penalty grows monotonically: averaged over six misspecification forms (linear, quadratic, cubic, absolute-value, sinusoidal, cross-term) F=1.00 gives $\Delta R^2 \approx 0$, F=0.38 gives ≈ -0.33, and an over-rotation stress test (F=−0.50) gives ≈ -2.0 (Fig. S8); all 24 family $\times$ form sweeps are individually monotone, and the ordering is identical across four unrelated teacher families (linear, monotone-nonlinear, kinetics-exponential, PDE-field), whose curves are visually coincident. That coincidence says the teacher factor carries no information *in this construction*; it is not evidence that the effect is domain-general, and the panel does not resolve the six misspecification forms, which do spread (Fig. S8). Placing the real system on the sweep only illustratively (the real-closure-to-F map is a synthetic construction, not a calibrated measurement), the COSMO-SAC closure over predicted σ-profiles falls at $F \lesssim 0.4$: its solubility $\Delta R^2 \approx -0.36$ is what a closure of this fidelity predicts, consistent with B_{closure} dominating B_{insuff}. The two ΔR^2 values are measured on different targets—synthetic m on the sweep, $\ln x_2$ through the SLE solver and the crystal head in the real system—which is the other reason the placement carries no calibration.

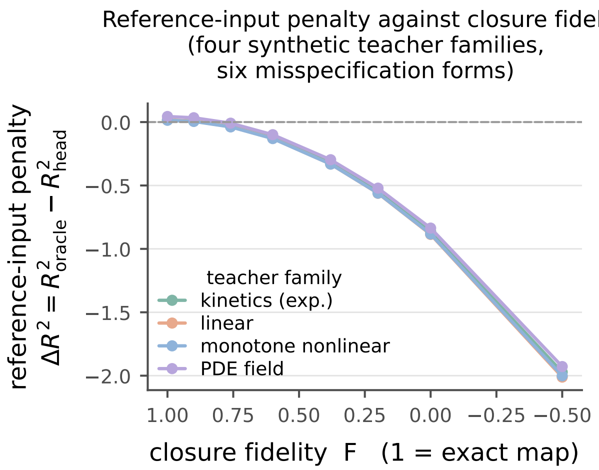

Figure S8: Closure-fidelity sweep on synthetic, non-chemical teacher functions. As the closure is made more misspecified (fidelity F falling from 1; axis reversed), the reference-input penalty $\Delta R^2 = R^2_{\text{oracle}} - R^2_{\text{head}}$ falls monotonically below zero. Each curve is one of four teacher families, averaged over six misspecification forms, which the panel does not resolve. No point is marked for the real closure. *Caveat:* here ΔR^2 *equals* B_{closure} by design, so the sweep is a check on the construction and not an independent test of the hypothesis; and the four curves are visually coincident because the teacher factor carries no information in this construction, not because the effect is domain-general. Scope: §S11.

The same construction shows why the grounding gap is an indicator, not a certificate (§2.1.2). Closure fidelity and input sufficiency are varied separately: when $z^\star$ is sufficient, $\mathcal{G}$ tracks B_{closure} as fidelity varies (Theorem S1's proxy is exact); when $z^\star$ is lossy but the closure is *well specified*, $B_{\text{closure}} = 0$ throughout yet $\mathcal{G} > 0$ grows with the discarded variance. A positive grounding gap can thus be pure input insufficiency, not closure error—so what closure dominance is claimed on rests on the law-of-total-variance bound of the decomposition, computed under the deployed convention (the bound is convention-specific even though the estimand it bounds is not; §2.1.1). That claim is made stratum by stratum and on one row set, the glycol-ether solvents, and through a one-sided bound; the aggregate does not stand (§3.2.1). $\mathcal{G}$ only corroborates it.

S12 Positioning against adjacent literatures

§1 places the instrument against the model-discrepancy problem of Kennedy and O'Hagan[S20] and its calibration critique,[S21] which is the frame the paper works in. Four further literatures bear on it without being that frame. Separating whether predictions are right on average from whether they are sharp is Murphy and DeGroot's reliability–resolution decomposition:[S22,S23] the closure/insufficiency split is an object of the same kind, an identity that partitions a squared error, but on a *fixed* operator with an external reference for the intermediate rather than on a forecast–outcome pair. A misspecified model still converges to a pseudo-true limit under White's quasi-maximum-likelihood account,[S16] which is what Δ_∞ of Proposition 1 names for a composed predictor. That a decomposition may be only weakly constrained by the data is the subject of parameter sloppiness[S24,S25] and of structural versus practical identifiability[S26,S27]—the setting of Lemma S1. That apparent skill may reflect pipeline choices rather than physics is the concern of underspecification,[S28] leakage[S29] and compositional risk,[S30] which is why every arm difference in this paper is reported as a difference between two separately tuned pipelines (§3.6). Finally, that a constrained prior lowers variance at the cost of a bias confined to the directions the data leave undetermined is the null-space reading of regularisation[S31,S32] and of null-space priors in inverse problems,[S33] which is the frame in which Proposition 1 is stated.

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